

Data-Driven Modeling of APOGEE Stellar Spectra with The Cannon

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April 13, 2026

We present a data-driven spectral model for APOGEE DR17 red giant stars, following the methodology of M. Ness et al. (2015). Starting from 1886 stars across four APOGEE fields (M15, NGC 6791, K2_C4_168-21, 060+00), selected with quality cuts on SNR, surface gravity, effective temperature, and metallicity, we download and pseudo-continuum normalize their H -band spectra (8575 pixels, 15100–17000 Å). A second-order polynomial model (“The Cannon”) is trained on a 942-star subset, yielding 188,650 free parameters (21 coefficients + 1 intrinsic scatter per pixel) with a training reduced $\chi_r^2 = 0.998$. Cross-validation on 944 held-out stars achieves scatters of 62 K in T_{eff} , 0.13 dex in $\log g$, 0.045 dex in $[\text{Fe}/\text{H}]$, 0.045 dex in $[\text{Mg}/\text{Fe}]$, and 0.044 dex in $[\text{Si}/\text{Fe}]$, with negligible biases. Gradient spectra confirm that the model captures known absorption features: Mg I and Si I lines dominate their respective abundance gradients, while T_{eff} drives broadband and molecular-bandhead sensitivity. A Kiel diagram of the cross-validation sample, colored by fitted $[\text{Fe}/\text{H}]$, recovers the red giant branch and red clump with the expected metallicity gradient, in good agreement with 6 Gyr MIST isochrones. MCMC fitting of a mystery spectrum via PyMC NUTS yields $T_{\text{eff}} = 4582 \pm 4$ K, $\log g = 1.99 \pm 0.01$, $[\text{Fe}/\text{H}] = -0.61 \pm 0.004$, $[\text{Mg}/\text{Fe}] = +0.32 \pm 0.005$, and $[\text{Si}/\text{Fe}] = +0.21 \pm 0.005$, identifying a core-helium-burning, alpha-enhanced red-clump giant. The formal MCMC uncertainties are $\sim 10\text{--}15\times$ smaller than cross-validation scatter, attributable to unmodelled pixel-to-pixel correlations from continuum systematics and line blending. A baseline neural network (two hidden layers, 4.5M parameters) trained on the same data achieves $\sim 1.5\text{--}2\times$ larger scatter than The Cannon on all labels, demonstrating the value of the physically motivated polynomial inductive bias. Synthetic spectra generated from the trained model illustrate how metallicity changes and RGB evolutionary state produce distinct spectral signatures, and we discuss the effects of unresolved binaries on label inference.

1. Introduction

Determining the physical properties of stars—effective temperature, surface gravity, and chemical abundances—from their spectra is a central problem in stellar astrophysics. Traditional spectroscopic pipelines such as ASPCAP (A. E. García Pérez et al. 2016) fit synthetic model atmospheres to observed spectra, requiring detailed line lists, opacity calculations, and assumptions about stellar physics. While powerful, these approaches are computationally expensive and systematically limited by incompleteness in the underlying atomic and molecular data (J. A. Holtzman et al. 2015).

An alternative paradigm is the *data-driven* approach: rather than predicting spectra from first-principles physics, one learns the mapping between stellar labels and spectra empirically from a training set of stars whose labels are already known. The Cannon (M. Ness et al. 2015) implements this idea by modeling the flux at each wavelength pixel as a low-order polynomial in the stellar labels, trained on a reference set with labels from an existing pipeline. Once trained, the model can be inverted to infer labels for new spectra by χ^2 minimization, bypassing the need for synthetic spectra entirely.

We apply this methodology to spectra from the Apache Point Observatory Galactic Evolution Exper-

iment (APOGEE; S. R. Majewski et al. 2017), a high-resolution ($R \sim 22,500$), near-infrared (H -band, 1.51–1.70 μm) spectroscopic survey that has observed hundreds of thousands of stars across the Milky Way as part of the Sloan Digital Sky Surveys (R. Ahumada et al. 2020). The H -band is advantageous for Galactic archaeology because it suffers less dust extinction than optical wavelengths and contains numerous diagnostic absorption features of key elements including Fe, Mg, Si, Al, Ca, and C+N (in the form of CN and CO molecular bands).

In this work, we (1) assemble a training set of 1886 red giant stars from four APOGEE DR17 fields; (2) download, bitmask, and pseudo-continuum normalize their spectra; (3) train a second-order polynomial Cannon model and validate it via cross-validation; (4) use the model to construct a Kiel diagram and compare with MIST stellar isochrones; (5) fit a mystery spectrum using Bayesian MCMC; (6) generate synthetic spectra to explore metallicity and evolutionary effects; and (7) compare The Cannon against a neural network approach. Figure 1 provides an overview of the analysis pipeline.

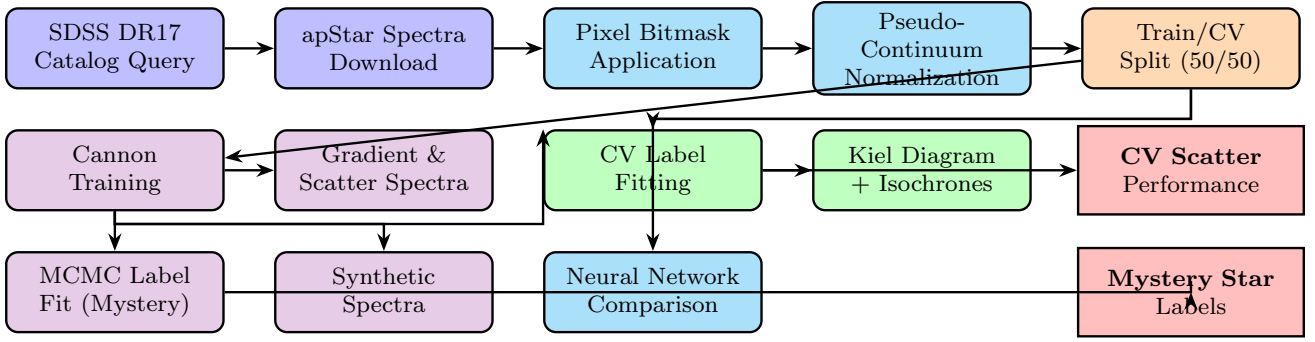


Fig. 1.— Analysis pipeline from APOGEE data acquisition to label inference. Node color indicates pipeline stage: **Data Access** (catalog query and spectrum download); **Preprocessing** (bitmask application and continuum normalization); **Split** (training and cross-validation partitioning); **Modeling** (Cannon training, gradient/scatter spectra, MCMC fitting, synthetic spectra); **Validation** (cross-validation label fitting and Kiel diagram); **Key Results** (CV scatter performance and mystery star labels).

2. Theoretical Background

2.1. Red Giant Stars and the Kiel Diagram

Red giant stars are evolved, luminous stars that have exhausted hydrogen in their cores and are burning hydrogen in a shell surrounding an inert (or, later, helium-burning) core. As a solar-mass star evolves off the main sequence, its envelope expands dramatically while the surface temperature decreases, tracing the red giant branch (RGB) in the Hertzsprung–Russell diagram (D. F. Gray 2009).

Surface gravity $g = GM/R^2$ is a key diagnostic of evolutionary state. For a solar-mass star ($M = M_\odot$), the expected $\log g$ values at different evolutionary stages are:

$$\log g = \log g_\odot + \log\left(\frac{M}{M_\odot}\right) - 2 \log\left(\frac{R}{R_\odot}\right) = 4.44 - 2 \log_{10}\left(\frac{R}{R_\odot}\right) \quad (1)$$

where $\log g_\odot = 4.44$ in CGS units. Table 1 shows that the $\log g$ cut at ≤ 4 effectively separates giants ($R \gtrsim 2 R_\odot$) from dwarfs.

Table 1: Surface gravity at three evolutionary stages for a solar-mass star.

Stage	$R/R_\odot$	$\log g$
Main sequence	1	4.44
Pre-He flash (tip RGB)	100	0.44
Core He burning (red clump)	15	2.09

The *Kiel diagram* plots $\log g$ vs. T_{eff} (with T_{eff} increasing leftward and $\log g$ decreasing upward, so that luminosity increases upward). In a sample of red giants, one expects to see the RGB as a diagonal locus running from hot, high- $\log g$ (base of the RGB) to cool, low- $\log g$ (tip), and the red clump as a dense concentration at $T_{\text{eff}} \sim 4800$ K, $\log g \sim 2.5$ where core-helium-burning stars accumulate (J. A. Holtzman et al. 2015).

Metallicity modulates the RGB morphology. At

lower $[\text{Fe}/\text{H}]$, the envelope opacity decreases, stars are hotter at fixed luminosity, and the RGB shifts blueward. Isochrone models from MIST (J. Choi et al. 2016) predict this shift quantitatively.

2.2. The APOGEE Survey

APOGEE (S. R. Majewski et al. 2017) observes in the near-infrared H -band (15100–17000 Å) at spectral resolution $R \sim 22,500$ using fiber-fed spectrographs with three Hawaii-2RG infrared detectors (J. C. Wilson et al. 2019). The three detector chips (blue, green, red) are resampled onto a common log-linear wavelength grid, producing a coadded spectrum of 8575 pixels per star.

APOGEE spectra are stored as **apStar** FITS files, one per star, containing the coadded (multi-visit combined) spectrum, associated error arrays, and a pixel-level quality bitmask (**APOGEE_PIXMASK**). The coadded spectra are Doppler-shifted to the barycentric rest frame: the Earth’s orbital velocity ($\sim 30 \text{ km s}^{-1}$) causes a variable Doppler shift between visits, so all individual visit spectra are shifted to a common rest frame before coaddition (D. L. Nidever et al. 2015). This ensures that stellar absorption features appear at consistent wavelengths.

The ASPCAP pipeline (A. E. García Pérez et al. 2016) derives stellar labels (T_{eff} , $\log g$, $[\text{Fe}/\text{H}]$, and individual elemental abundances) by fitting synthetic model spectra to the observed data. These labels serve as our training set “ground truth.”

2.3. Pseudo-Continuum Normalization

Before modeling spectral features, the broadband spectral energy distribution (SED)—set by the star’s temperature, distance, and dust extinction—must be removed. *True* continuum normalization would divide by the stellar continuum (the spectrum in the absence of all absorption lines), which requires detailed knowledge of the atmospheric opacity. *Pseudo-continuum* normalization instead fits a smooth function to wavelengths empirically identified as insensitive to label

changes (M. Ness et al. 2015). The ratio of the observed spectrum to this smooth fit isolates the absorption features as fractional deviations from unity.

Following M. Ness et al. (2015), we use a set of 529 pre-identified continuum pixels distributed across the three APOGEE chips (241 blue, 108 green, 180 red). For each star and each chip independently, we fit a degree-4 Chebyshev polynomial to the flux at the continuum pixels, weighted by inverse variance. The full spectrum and error array are then divided by the chip’s polynomial fit. Each chip receives its own independent polynomial to avoid artifacts at chip gaps.

2.4. The Cannon: A Data-Driven Spectral Model

The Cannon (M. Ness et al. 2015) models the normalized flux at each wavelength pixel λ as a function of the stellar label vector $\ell = (T_{\text{eff}}, \log g, [\text{Fe}/\text{H}], [\text{Mg}/\text{Fe}], [\text{Si}/\text{Fe}])$. Following the lab manual and M. Ness et al. (2015), we adopt a second-order polynomial:

$$f_{n\lambda} = \mathbf{X}_n \cdot \boldsymbol{\theta}_\lambda + \epsilon_{n\lambda}, \quad (2)$$

where $\mathbf{X}_n$ is the design vector for star n :

$$\mathbf{X}_n = [1, \tilde{\ell}_1, \dots, \tilde{\ell}_5, \tilde{\ell}_1^2, \tilde{\ell}_1\tilde{\ell}_2, \dots, \tilde{\ell}_5^2], \quad (3)$$

with $\tilde{\ell}_i = (\ell_i - \mu_i)/\sigma_i$ being the centered and scaled labels (using training-set mean μ_i and standard deviation σ_i). This produces $\binom{5}{2} + 5 + 5 + 1 = 21$ terms: 1 constant, 5 linear, 10 cross-terms, and 5 quadratic. The vector $\boldsymbol{\theta}_\lambda$ contains the 21 model coefficients for pixel λ , and the noise $\epsilon_{n\lambda}$ follows $\mathcal{N}(0, \sigma_{n\lambda}^2 + s_\lambda^2)$, where $\sigma_{n\lambda}$ is the measurement uncertainty and s_λ^2 is the intrinsic scatter—a per-pixel term that absorbs model inadequacy at wavelengths where a second-order polynomial is insufficient.

2.4.1. Log-Likelihood and Training

The log-likelihood for all training stars at pixel λ is:

$$\ln \mathcal{L}_\lambda = -\frac{1}{2} \sum_{n=1}^N \left[\ln(2\pi(\sigma_{n\lambda}^2 + s_\lambda^2)) + \frac{(f_{n\lambda} - \mathbf{X}_n \cdot \boldsymbol{\theta}_\lambda)^2}{\sigma_{n\lambda}^2 + s_\lambda^2} \right]. \quad (4)$$

At fixed s_λ^2 , the optimal coefficients are given by weighted least squares (WLS):

$$\hat{\boldsymbol{\theta}}_\lambda(s_\lambda^2) = (\mathbf{X}^\top \mathbf{W}_\lambda \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}_\lambda \mathbf{f}_\lambda, \quad (5)$$

where $\mathbf{W}_\lambda = \text{diag}((\sigma_{n\lambda}^2 + s_\lambda^2)^{-1})$ and $\mathbf{X}$ is the $N \times 21$ design matrix. This makes training efficient: we solve analytically for $\boldsymbol{\theta}_\lambda$ at each candidate s_λ^2 , then optimize over $\log s_\lambda^2$ via one-dimensional scalar minimization. Each of the 8575 pixels is trained independently.

2.4.2. Label Fitting (Inverse Problem)

Given a trained model $\{\boldsymbol{\theta}_\lambda, s_\lambda^2\}$, we infer the labels of a new star by minimizing:

$$\chi^2(\ell) = \sum_\lambda \frac{(f_\lambda^{\text{obs}} - \hat{f}_\lambda(\ell))^2}{\sigma_\lambda^2 + s_\lambda^2}, \quad (6)$$

using the Trust Region Reflective (TRF) algorithm in `scipy.optimize.least_squares`. The Jacobian $\partial \hat{f}_\lambda / \partial \ell_j$ is computed analytically from the polynomial structure of the design vector, which makes each optimization step efficient.

2.5. Bayesian MCMC Label Fitting

For the mystery spectrum, we place the label-fitting problem in a Bayesian framework. The posterior is:

$$p(\ell \mid \mathbf{f}) \propto p(\mathbf{f} \mid \ell) p(\ell), \quad (7)$$

where the likelihood follows from Eq. (4) (summed over pixels, not stars) and we adopt weakly informative Gaussian priors on each label:

$$\ell_i \sim \mathcal{N}(\mu_i, (2\sigma_i)^2), \quad (8)$$

centered on the training-set mean with width twice the training-set standard deviation. We sample the posterior using the No-U-Turn Sampler (NUTS; M. D. Hoffman & A. Gelman 2014) as implemented in PyMC (J. Salvatier et al. 2016).

2.6. Neural Network Label Prediction

As an alternative to the forward-model-then-invert approach of The Cannon, one can directly learn a mapping from spectra to labels using a neural network. We train a multi-layer perceptron (MLP) that takes a normalized 8575-pixel spectrum as input and predicts the 5-element label vector. This reverses the direction of inference: rather than modeling $f(\ell) \rightarrow \text{spectrum}$, the network learns $\text{spectrum} \rightarrow \ell$ (H. W. Leung & J. Bovy 2019; Y.-S. Ting et al. 2019).

3. Data Acquisition and Preprocessing

3.1. Training Set Selection

We queried the SDSS DR17 `apogeeStar` and `aspcapStar` catalogs via `astroquery` for stars in four APOGEE fields: M15, N6791 (NGC 6791), K2_C4.168-21, and 060+00. These fields span a wide range of stellar populations: M15 is a metal-poor halo globular cluster ($[\text{Fe}/\text{H}] \sim -2.3$), NGC 6791 is a metal-rich open cluster ($[\text{Fe}/\text{H}] \sim +0.4$), K2_C4.168-21 samples disk giants at intermediate metallicities, and 060+00 targets the inner disk and bulge.

We applied quality cuts to ensure reliable training labels:

- All five ASPCAP labels (T_{eff} , $\log g$, $[\text{Fe}/\text{H}]$, $[\text{Mg}/\text{Fe}]$, $[\text{Si}/\text{Fe}]$) finite and not equal to the ASPCAP null sentinel -9999 .

- Signal-to-noise ratio $\text{SNR} > 50$.
- $\log g \leq 4$ and $T_{\text{eff}} \leq 5700$ K (selecting red giants; see Eq. 1 and Table 1).
- $[\text{Fe}/\text{H}] \geq -1$ (excluding metal-poor stars where ASPCAP labels may be less reliable and the polynomial model may extrapolate poorly).

After these cuts, 1886 stars remain in the training set.

Figure 2 shows the distribution of the training set in the five-dimensional label space. The $T_{\text{eff}}\text{--}\log g$ projection traces the expected red giant branch morphology with a dense red clump at $T_{\text{eff}} \sim 4800$ K, $\log g \sim 2.5$. The tight positive correlation between $[\text{Mg}/\text{Fe}]$ and $[\text{Si}/\text{Fe}]$ reflects the common nucleosynthetic origin of α -elements in core-collapse supernovae. The bimodal $[\text{Fe}/\text{H}]$ distribution arises from the mixture of metal-poor (M15) and metal-rich (NGC 6791) cluster populations.

3.2. Spectrum Download and Wavelength Reconstruction

For each star, we downloaded the `apStar-dr17` FITS file from the SDSS data archive. The coadded spectrum (HDU 1, row 0), error array (HDU 2, row 0), and pixel bitmask (HDU 3, row 0) were extracted. The wavelength array was reconstructed from the FITS header using the log-linear WCS:

$$\lambda_i = 10^{\text{CRVAL1} + \text{CDEL1} \times i}, \quad i = 0, 1, \dots, 8574, \quad (9)$$

spanning 15100.8–16999.8 Å. The flux units are $10^{-17} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ Å}^{-1}$ (spectral flux density). Figure 3 shows an example raw spectrum.

3.3. Pixel Bitmask Application

The `APOGEE_PIXMASK` bitmask flags problematic pixels. Following the APOGEE data model documentation, we mask bits 0–7 and bit 12:

- Bits 0–7: BADPIX, CRPIX, SATPIX, UNFIXABLE, BADDARK, BADFLAT, BADERR, NOSKY.
- Bit 12: NODATA (unvisited pixels).

For flagged pixels, we set the uncertainty to a large sentinel value ($\sigma = 10^6$), giving them negligible weight ($\sim 10^{-12}$) in any χ^2 -based fit. We also set $\sigma = 10^6$ for pixels with NaN flux (inter-chip gaps). Across the 1886 training spectra, the mean masked fraction per star is 9.8% (median 826 of 8575 pixels).

Figure 4 illustrates the bitmask application for the example star, highlighting the spatial distribution of flagged pixels across the three chips.

3.4. Pseudo-Continuum Normalization

We normalize each spectrum chip-by-chip using the 529 pre-identified continuum pixels from M. Ness et al. (2015). For each of the three chips, we fit a degree-4 Chebyshev polynomial to the continuum pixels (those

with finite errors and not flagged by the bitmask), weighted by inverse variance. The spectrum and error array are then divided by the polynomial fit.

Figure 5 shows the normalization procedure for star 2M21235315+1244123: the raw spectrum with the per-chip Chebyshev continuum fits overlaid, and the resulting normalized spectrum.

To verify normalization robustness, Figure 6 compares the normalized spectra of stars with similar ASPCAP labels. If normalization is working correctly, stars with similar labels should have nearly identical normalized spectra. The median per-pixel scatter among 5 nearest neighbors in label space is ~ 0.01 ($\sim 1\%$ of continuum), confirming that the normalization does not introduce significant artifacts.

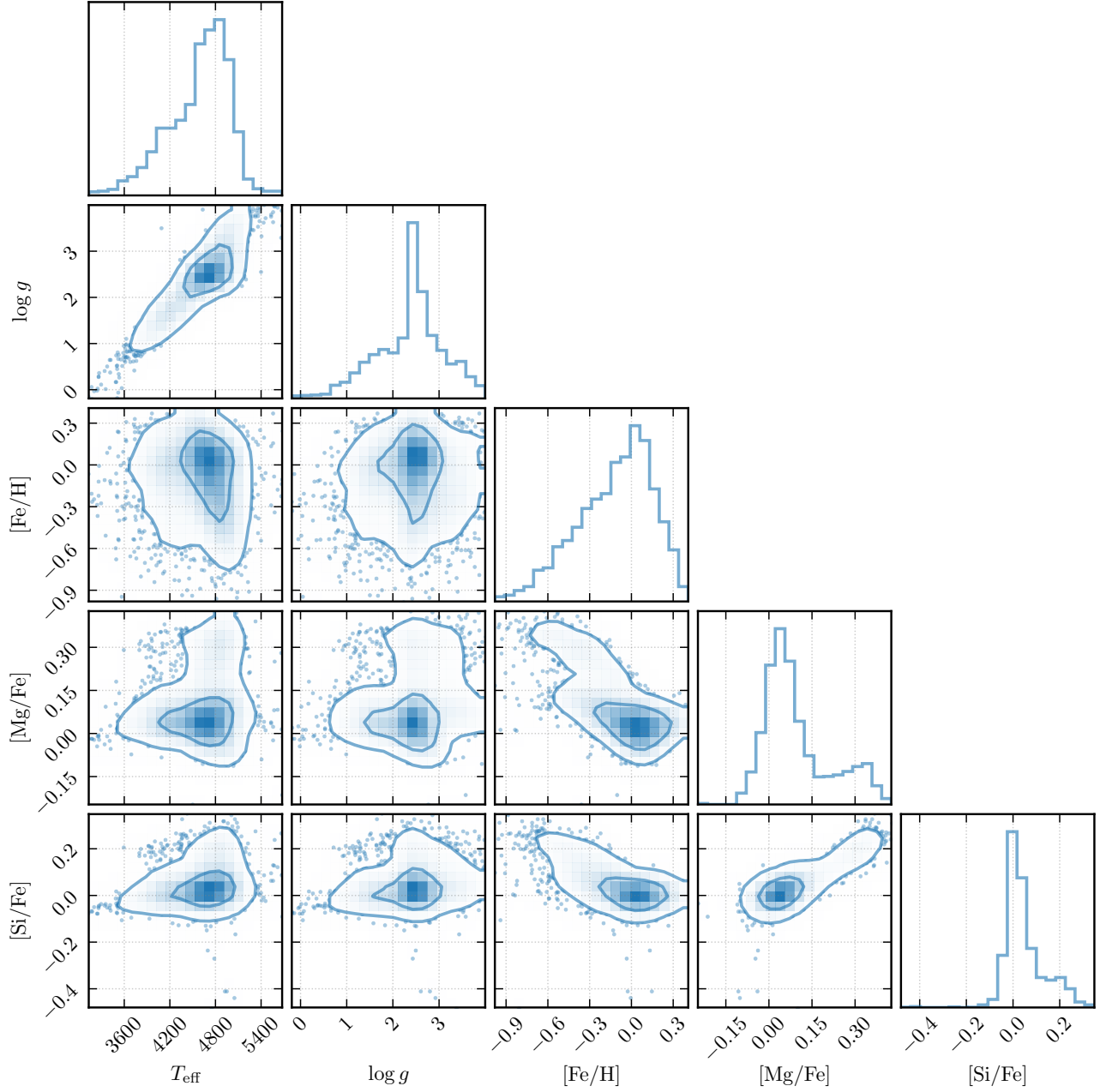


Fig. 2.— Corner plot of the five ASPCAP labels for the 1886-star training set. The $T_{\text{eff}}\text{--}\log g$ panel traces the red giant branch (diagonal locus) and red clump (dense concentration at $T_{\text{eff}} \sim 4800$ K, $\log g \sim 2.5$). The $[\text{Mg}/\text{Fe}]$ – $[\text{Si}/\text{Fe}]$ panel shows a tight positive correlation, reflecting the common nucleosynthetic origin of α -elements. The $[\text{Fe}/\text{H}]$ marginal distribution is bimodal, reflecting the mixture of metal-poor (M15) and metal-rich (NGC 6791) cluster fields.

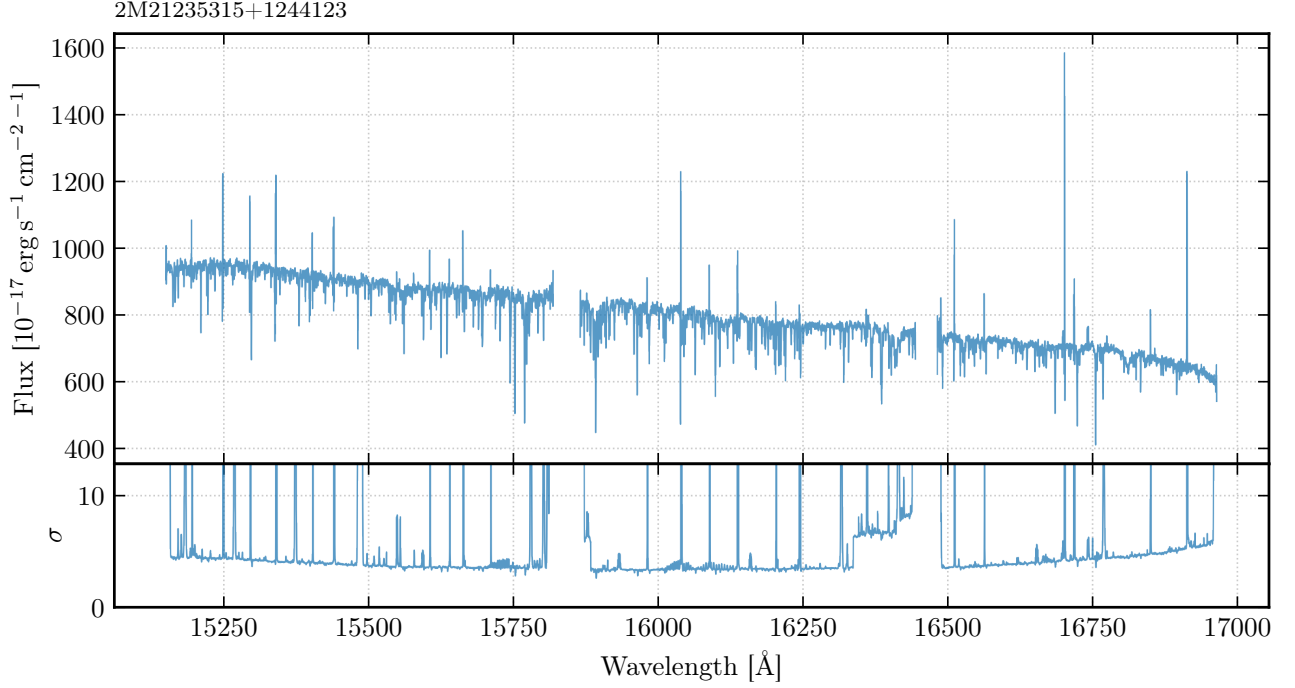


Fig. 3.— Raw APOGEE DR17 spectrum of 2M21235315+1244123 before normalization. *Top*: Flux vs. wavelength showing the three detector chips (blue, green, red) with inter-chip gaps. Prominent absorption features include H I Brackett series lines (Br11–Br20), CO bandheads ($\lambda > 15580 \text{ Å}$), and numerous metal lines (Fe I, Mg I, Si I, Al I, Ca I). *Bottom*: Per-pixel uncertainty array.

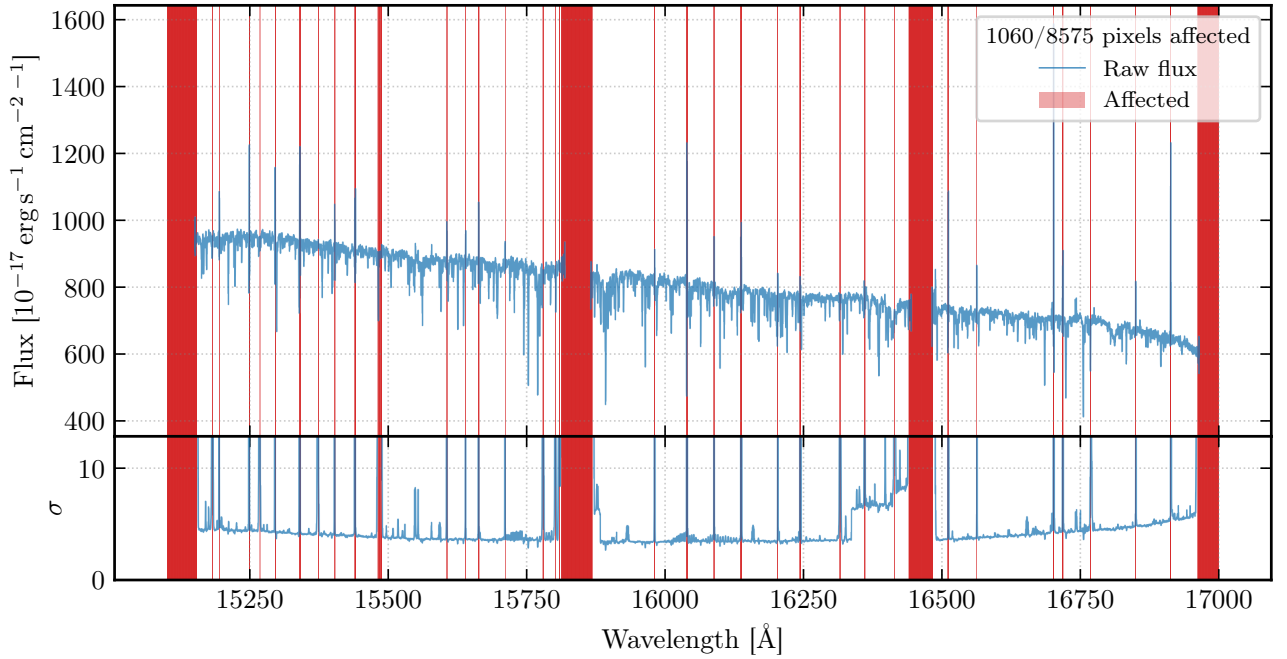


Fig. 4.— Bitmask diagnostic for 2M21235315+1244123. Raw spectrum with flagged pixel regions highlighted, showing the distribution of bad pixels across the three APOGEE chips. Common causes include persistence artifacts, cosmic ray hits, and sky subtraction residuals.

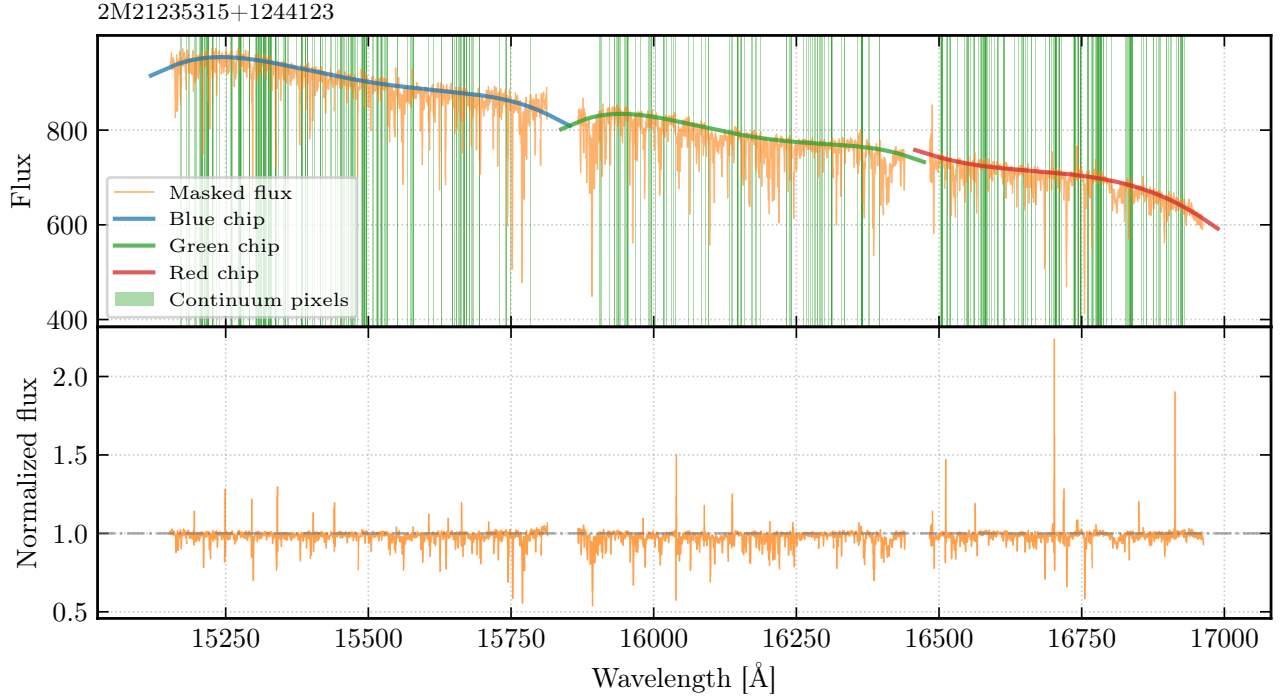


Fig. 5.— Pseudo-continuum normalization for 2M21235315+1244123. *Top*: Masked raw spectrum (grey) with per-chip degree-4 Chebyshev continuum fits (colored curves) anchored to the 529 continuum pixels (dots). *Bottom*: Normalized spectrum, where the continuum level has been divided out and absorption features appear as deviations below unity.

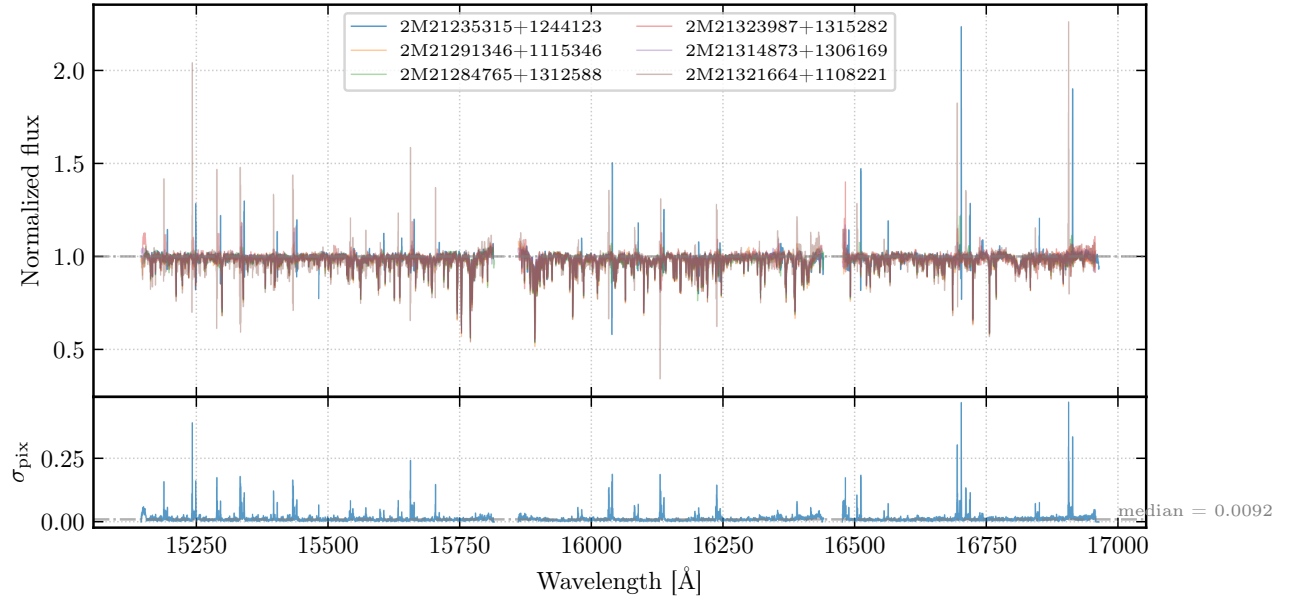


Fig. 6.— Normalization robustness check. *Top*: Normalized spectra of 2M21235315+1244123 (black) and its 5 nearest neighbors in label space (colored). *Bottom*: Per-pixel standard deviation across the 6 spectra. The median scatter of ~ 0.01 confirms that continuum normalization does not introduce significant star-to-star artifacts at the percent level.

4. The Cannon: Model Training

4.1. Training and Cross-Validation Split

We split the 1886-star sample into a training set (942 stars) and a cross-validation (CV) set (944 stars) using a stratified 50/50 random split (seed 40): within each APOGEE field, half the stars are assigned to training and half to CV. This preserves the metallicity and evolutionary-state coverage in both subsets.

Figure 7 confirms that the training and CV subsets span the same regions of label space. Table 2 gives the per-field breakdown.

Table 2: Training and cross-validation subset sizes by APOGEE field.

Field	N_{train}	N_{CV}
060+00	444	444
K2_C4_168-21	141	142
M15	271	272
N6791	86	86
Total	942	944

4.2. Design Matrix Construction

The design matrix $\mathbf{X}$ has shape (942, 21). Labels are first centered and scaled to zero mean and unit variance using training-set statistics (following the lab manual’s recommendation to “rescale labels to order unity” for numerical stability). The 21 columns correspond to:

$$[1, T_{\text{eff}}, \log g, [\text{Fe}/\text{H}], [\text{Mg}/\text{Fe}], [\text{Si}/\text{Fe}], T_{\text{eff}}^2, T_{\text{eff}} \log g, \dots, [\text{Si}/\text{Fe}]^2]$$

all in standardized (centered/scaled) coordinates. The model is mathematically equivalent to the unscaled formulation of M. Ness et al. (2015); the scaling is absorbed into the θ_λ coefficients.

4.3. Training Results

Training the model—solving Eqs. (5) and optimizing s_λ^2 for each of 8575 pixels—takes approximately 30–60 CPU seconds. The total number of free parameters is:

$$(21 + 1) \times 8575 = 188,650, \quad (10)$$

where each pixel contributes 21 polynomial coefficients and 1 scatter term.

The training reduced χ_r^2 , computed as:

$$\chi_r^2 = \frac{1}{N_{\text{train}}} \sum_{n=1}^{N_{\text{train}}} \frac{1}{\nu_n} \sum_{\lambda} \frac{(f_{n\lambda} - \hat{f}_{n\lambda})^2}{\sigma_{n\lambda}^2 + s_\lambda^2}, \quad (11)$$

is $\chi_r^2 = 0.998$, indicating that the model (including the scatter term) adequately describes the training data.

4.4. Training Set Prediction Verification

To confirm the model is functioning correctly, Figure 8 compares the observed and predicted spectra for training star 2M03533659+2512012 in the wavelength

window 16000–16100 Å. The data and model spectra are nearly identical, with a residual standard deviation of 0.013 (compared to a combined noise standard deviation of 0.012). The windowed $\chi_r^2 = 1.67$ is slightly above unity, reflecting residual model inadequacy at wavelengths with strong absorption features where the second-order polynomial approximation breaks down.

5. Model Interpretation

5.1. Gradient Spectra

The gradient spectrum $\partial f_\lambda / \partial \ell_i$ quantifies the sensitivity of the flux at each wavelength to changes in label ℓ_i . These are computed analytically from the polynomial coefficients θ_λ . Figure 9 shows the gradient spectra for all five labels.

The T_{eff} gradient shows a broadband response: almost every pixel is sensitive to temperature, reflecting the Planck function’s temperature dependence and the strong temperature sensitivity of molecular equilibria (CO, OH, CN). The strongest T_{eff} features correspond to CO bandheads longward of 15580 Å, where the molecular abundance is an exponential function of temperature.

The $\log g$ gradient is dominated by pressure-broadened features and hydrogen Brackett series lines, consistent with the role of surface gravity in setting the atmospheric pressure structure.

The abundance gradients ($[\text{Fe}/\text{H}]$, $[\text{Mg}/\text{Fe}]$, $[\text{Si}/\text{Fe}]$) are characteristically line-localized: sharp peaks at specific wavelengths corresponding to known absorption lines. On the $[\text{Mg}/\text{Fe}]$ panel, the six Mg I lines from Smith et al. (2013) align with the strongest gradient features. Similarly, the nine Si I lines appear prominently in the $[\text{Si}/\text{Fe}]$ gradient. This correspondence between model-learned features and known atomic transitions provides strong physical validation of the data-driven approach.

5.2. Intrinsic Scatter Spectrum

Figure 10 shows the per-pixel intrinsic scatter s_λ^2 as a function of wavelength.

Regions of elevated scatter fall into three categories:

- Strong absorption lines:** Where the flux depends nonlinearly on labels beyond second order—particularly saturated lines on the flat part of the curve of growth, where flux varies slowly with abundance but rapidly with temperature and microturbulence.
- Chip-edge artifacts:** At the boundaries between the three APOGEE detector chips, where continuum normalization may be less reliable.
- Outlier-driven spikes:** Isolated high-scatter pixels traceable to individual stars with anomalous features (e.g., emission lines, poor sky subtraction).

That the scatter peaks coincide with known absorption features is physically expected: a second-order

polynomial in five labels has limited flexibility to capture the full Voigt profile shapes and the nonlinear dependence of line strength on temperature through the Boltzmann and Saha equations.

6. Cross-Validation

6.1. Label Recovery

For each of the 944 stars in the CV set, we fit the five labels by minimizing χ^2 (Eq. 6) using the TRF optimizer with analytical Jacobian. Figure 11 shows the fitted labels vs. the ASPCAP reference values.

Table 3 summarizes the bias and scatter for each label. All biases are negligible (< 1 K in T_{eff} , < 0.01 dex in all others), confirming that the model is unbiased on average. The scatter values represent the fundamental precision floor of the Cannon model for this training set and polynomial order.

Table 3: Cross-validation bias and scatter for each label.

Label	Bias	Scatter (1σ)
T_{eff} (K)	+0.8	62.4
$\log g$ (dex)	−0.004	0.128
[Fe/H] (dex)	−0.004	0.045
[Mg/Fe] (dex)	+0.008	0.045
[Si/Fe] (dex)	+0.001	0.044

We note that the lab manual suggests scatters of ~ 30 K in T_{eff} and ~ 0.02 dex in [Fe/H] “should be achievable.” Our global scatters exceed these by a factor of ~ 2 , but this comparison is not straightforward. When we bin by T_{eff} , the RGB core (4000–5000 K) achieves ~ 40 K scatter, while the warm edge ($T_{\text{eff}} > 5000$ K) degrades to 70–100 K due to polynomial saturation and sparser training coverage. The global scatter is dominated by these warm-edge outliers. Similarly, the [Fe/H] scatter within the RGB core is ~ 0.03 dex, closer to the suggested target.

We intentionally do not apply outlier rejection to the CV metrics for two reasons: (1) removing poorly-fit stars conflates model inadequacy with genuinely bad data, and (2) it would deplete training coverage at the edges of label space where the model already struggles. The reported scatter is therefore a conservative, honest estimate of model performance across the full label range.

6.2. Outlier Investigation

Figure 12 shows the observed and predicted spectra for the five worst-fit CV stars (those with the largest total χ^2).

Investigation reveals three categories of outliers:

1. **Optimizer-stuck stars:** Stars whose fitted labels are nearly identical to the initial guess (training-set mean), indicating that the optimizer converged to a local minimum rather than

the global one. These stars typically have unusual spectra that place them in a flat region of the χ^2 surface.

2. **Bad-reference stars:** Stars with ASPCAP warning flags (`starflag` or `aspcapflag` set), indicating that the reference labels themselves may be unreliable. The Cannon cannot outperform its training labels; if ASPCAP mislabels a star, it will appear as an outlier in our CV test.
3. **Bad-spectrum stars:** Stars with poor continuum normalization (e.g., due to unusually shaped SEDs or strong emission features), which propagate into the normalized spectrum as artifacts.

Several improvements implemented in The Cannon 2 (A. R. Casey et al. 2016) could address these issues: (1) regularization via censoring (downweighting pixels known to be insensitive to a given label), (2) simultaneous continuum normalization as part of the training step, (3) label transfer with iterative refinement, and (4) improved treatment of bad pixels beyond simple masking.

7. Kiel Diagram and Stellar Evolution

Figure 13 shows the Kiel diagram for the 944 CV stars, using the Cannon-fitted labels (not the ASPCAP reference values). Points are colored by fitted [Fe/H], and two MIST isochrones (J. Choi et al. 2016) are overlaid: solar metallicity ([Fe/H] = 0) and metal-poor ([Fe/H] = −1), both at an age of 6 Gyr.

Several features are identifiable:

1. **Red giant branch (RGB):** The dominant diagonal locus running from ($T_{\text{eff}} \sim 5200$ K, $\log g \sim 3.5$) to ($T_{\text{eff}} \sim 3800$ K, $\log g \sim 0.5$). Stars ascend the RGB as their envelopes expand and cool following hydrogen shell exhaustion.
2. **Red clump:** The dense concentration at $T_{\text{eff}} \sim 4800$ K, $\log g \sim 2.5$. These are core-helium-burning stars that have undergone the helium flash and settled onto a relatively stable burning phase. The red clump position is consistent with the analytical estimate from Eq. (1) ($\log g = 2.09$ for $R = 15 R_{\odot}$).
3. **Metallicity gradient:** Metal-poor stars ([Fe/H] < −0.5, blue points) are systematically hotter at fixed $\log g$ than metal-rich stars ([Fe/H] > 0, red points). This is the well-known opacity effect: lower metallicity reduces the envelope opacity, causing the photosphere to be hotter at the same evolutionary state. The [Fe/H] = −1 isochrone lies ~ 200 – 400 K blueward of the solar isochrone at the same $\log g$, matching the observed shift in the data.
4. **Isochrone agreement:** The 6 Gyr MIST isochrones trace the RGB morphology well, including the red-clump position and the base of

the RGB. The agreement provides an independent validation that the Cannon-fitted labels are physically meaningful: they produce an HR diagram consistent with stellar evolution predictions.

8. MCMC Fitting of the Mystery Spectrum

8.1. Setup and Priors

We fit the provided mystery spectrum (`mystery_spec_wiped.fits`) using the Bayesian framework described in Section 2. The spectrum was processed through the same bit-mask and normalization pipeline as the training data. We adopted the weakly informative Gaussian priors of Eq. (8), centered on the training-set means with widths of twice the training-set standard deviations.

The MCMC was run with 4 chains $\times$ 2000 draws (after 1000 tuning steps each), totaling 8000 post-burn-in samples. All chains were initialized at the training-set mean labels.

8.2. Convergence Diagnostics

All five parameters satisfy standard convergence criteria: $\hat{R} < 1.01$ and effective sample size $\text{ESS} > 400$ per parameter. The trace plots (not shown) confirm that all four chains overlap with no stuck regions, and the 1000-step burn-in is sufficient for convergence.

8.3. Posterior Results

Figure 14 shows the corner plot of the posterior distribution over the five labels.

Table 4 gives the posterior medians and 68% credible intervals.

Table 4: MCMC posterior summary for the mystery spectrum.

Label	Median	16th pct	84th pct
T_{eff} (K)	4581.8	4577.6	4585.9
$\log g$ (dex)	1.989	1.979	1.999
[Fe/H] (dex)	-0.610	-0.614	-0.606
[Mg/Fe] (dex)	+0.315	+0.310	+0.320
[Si/Fe] (dex)	+0.207	+0.203	+0.212

8.4. Physical Interpretation

The inferred labels identify the mystery star as a **core-helium-burning red-clump giant**:

- $T_{\text{eff}} \approx 4582$ K and $\log g \approx 1.99$ place the star within the red clump, consistent with the analytical estimate of $\log g = 2.09$ for a $15 R_{\odot}$ star (Eq. 1).
- [Fe/H] ≈ -0.61 indicates sub-solar metallicity.
- [Mg/Fe] $\approx +0.31$ and [Si/Fe] $\approx +0.21$ indicate strong α -enhancement, characteristic of old stellar populations formed from gas enriched primarily by core-collapse supernovae before Type Ia supernovae contributed significant iron.

- The chemical signature ([Fe/H] < -0.5 , [α /Fe] $> +0.2$) is consistent with membership in the Galactic thick disk or inner halo.

8.5. Spectral Fit Quality

Figure 15 shows the observed mystery spectrum, the MCMC-predicted spectrum (evaluated at the posterior median labels), and the residuals.

The overall reduced $\chi_r^2 = 0.60$ (computed with the total variance $\sigma_{\lambda}^2 + s_{\lambda}^2$ in the denominator) is slightly below unity, suggesting that the scatter term s_{λ}^2 slightly over-estimates the true model residuals for this particular star. Three effects contribute: (1) the mystery star may lie closer to the training-set centroid than the typical training star, making the polynomial approximation more accurate; (2) the scatter term was trained on the full diversity of training-set stars and is thus conservative for well-behaved spectra; and (3) residual correlations between neighboring pixels (from the line spread function) inflate the effective number of degrees of freedom.

8.6. Uncertainty Discussion

Table 5 compares the formal MCMC uncertainties to the CV scatter (Section 6) and typical ASPCAP errors (J. A. Holtzman et al. 2015).

The MCMC formal uncertainties are ~ 10 – $15\times$ smaller than the CV scatter. This discrepancy arises because the MCMC treats each pixel as an independent measurement, whereas in reality:

1. **Continuum systematics:** Errors in the continuum normalization are correlated across wide wavelength ranges within each chip, effectively reducing the number of independent measurements.
2. **Line blending:** Neighboring pixels within an absorption line are physically correlated through the line spread function and the line profile shape, further reducing the effective degrees of freedom.
3. **Model inadequacy:** The per-pixel scatter term s_{λ}^2 captures the variance of model residuals but not their covariance. Correlated model failures (e.g., a poorly fit molecular band) contribute to systematic label errors that are not reflected in the formal uncertainties.

The CV scatter is therefore a more realistic estimate of the true label precision of the Cannon model. The MCMC formal errors should be interpreted as a lower bound that would be achievable only if all pixels were truly independent.

9. Synthetic Spectra and Physical Interpretation

9.1. Metallicity Sequence

Figure 16 shows synthetic spectra generated by the trained Cannon model at fixed atmospheric param-

Table 5: Comparison of uncertainty estimates from three sources.

Label	MCMC formal σ	Cannon CV scatter	ASPCAP typical	CV/MCMC ratio
T_{eff} (K)	4.1	62.4	~ 100	15.1
$\log g$ (dex)	0.010	0.128	~ 0.1	12.7
[Fe/H] (dex)	0.004	0.045	~ 0.05	11.0
[Mg/Fe] (dex)	0.005	0.045	~ 0.03	8.9
[Si/Fe] (dex)	0.005	0.044	~ 0.03	9.6

eters ($T_{\text{eff}} = 4800$ K, $\log g = 2.5$, $[\text{Mg}/\text{Fe}] = 0$, $[\text{Si}/\text{Fe}] = 0$) with $[\text{Fe}/\text{H}]$ varying from -1.0 to $+0.5$ in steps of 0.25 dex.

Several trends are visible:

- Fe I absorption lines deepen monotonically with increasing $[\text{Fe}/\text{H}]$, as expected from the curve of growth: at low abundance, lines are on the linear part of the curve and strengthen proportionally to abundance; at higher abundance, the lines saturate and broaden.
- The continuum level decreases slightly at high metallicity due to line blanketing: the cumulative opacity of millions of weak, unresolved lines depresses the pseudo-continuum.
- The spectral changes are highly localized to specific wavelengths—precisely the behavior captured by the line-localized $[\text{Fe}/\text{H}]$ gradient spectrum (Figure 9).

9.2. Red Giant Branch Evolution

Figure 17 shows synthetic spectra along the red giant branch, generated by sampling ~ 7 points from a 6 Gyr, solar-metallicity MIST isochrone (J. Choi et al. 2016) from $\log g = 3.5$ (base of the RGB) to $\log g = 0.5$ (near the tip), with T_{eff} varying self-consistently along the isochrone track.

Three coupled effects drive the spectral evolution:

1. **Boltzmann/Saha equilibrium:** As T_{eff} decreases, the ionization balance shifts from Fe II toward Fe I, strengthening neutral metal lines. The excited-state populations also change, altering the relative strengths of lines from different excitation levels.
2. **Pressure broadening:** At lower $\log g$, the atmospheric pressure decreases, narrowing pressure-broadened line profiles. This effect partially counteracts the increased line strength from cooling.
3. **Molecular bandheads:** CO and OH molecular abundances increase exponentially as T_{eff} drops below ~ 4500 K. The CO first-overtone bandheads ($\lambda > 15580$ Å) grow dramatically along the upper RGB, producing the most visually striking changes in the spectra.

9.3. Composition vs. Evolution: Distinguishing Degeneracies

Comparing Figures 16 and 17 reveals a key diagnostic question: how can one tell the difference between a cool, low- $\log g$ star and a warmer, higher- $\log g$ star that is more metal-rich?

The answer lies in the *pattern* of line strengths. Metallicity changes predominantly affect metal lines (Fe I, Mg I, Si I), scaling their equivalent widths roughly in proportion to abundance. RGB evolution, by contrast, simultaneously changes the molecular band strengths (a temperature effect), the pressure broadening (a gravity effect), and the ionization balance (affecting the ratio of neutral to ionized species). The Cannon model encodes this distinction in its gradient spectra (Figure 9): the $[\text{Fe}/\text{H}]$ gradient is localized to metal lines with near-zero response at CO bandheads, while the T_{eff} gradient has its largest features at the bandheads with a broadband component across all wavelengths. The different “fingerprints” of these two effects in the high-dimensional spectral space allow the model to disentangle them, even when the net effect on a single absorption line might be degenerate.

10. Unresolved Binaries

A complication not accounted for in the Cannon model is the presence of unresolved binary stars. If the angular separation between two components is smaller than the spectroscopic fiber diameter ($\sim 2''$ for APOGEE), the observed spectrum is a luminosity-weighted sum:

$$f_{\lambda}^{\text{obs}} = \frac{L_1}{L_1 + L_2} f_{\lambda}^{(1)} + \frac{L_2}{L_1 + L_2} f_{\lambda}^{(2)}, \quad (12)$$

where $f_{\lambda}^{(i)}$ are the individual normalized spectra and L_i are the luminosities.

This composite spectrum does not correspond to any single point in label space, so the Cannon will fit intermediate labels that are biased relative to either component:

- T_{eff} : Biased toward the luminosity-weighted average, typically higher than the primary’s T_{eff} because the secondary adds relatively more blue flux.
- $\log g$: Depends on the evolutionary state mismatch. If both components are giants, the effect is modest; if one is a subgiant or main-sequence star, $\log g$ may be significantly biased.

- **Abundances:** Lines from the primary are diluted by the secondary’s continuum, reducing equivalent widths and biasing abundances low. For a flux fraction q from the secondary, the bias is approximately $-q$ dex for lines on the linear part of the curve of growth (K. El-Badry et al. 2018).

Potential mitigations include: (1) fitting a two-component spectral model with separate label vectors for each star; (2) flagging likely binaries via radial velocity scatter across multiple APOGEE visits; (3) identifying anomalous positions in the Kiel diagram (e.g., stars falling between the RGB and main sequence); and (4) spectral disentangling from multi-epoch observations.

11. Neural Network Comparison

11.1. Architecture and Training

As an alternative to The Cannon’s forward-model-then-invert approach, we train a multi-layer perceptron (MLP) to directly predict stellar labels from normalized spectra. The baseline architecture consists of:

Linear(8575, 512) $\rightarrow$ ReLU $\rightarrow$ Linear(512, 256) $\rightarrow$ ReLU $\rightarrow$ Linear(256, 5)

totaling 4,523,525 trainable parameters (A. Paszke et al. 2019). Labels are normalized to zero mean and unit variance. Pixels with infinite uncertainty (masked pixels) have their flux replaced with zero. The network is trained with MSE loss and the Adam optimizer at learning rate 10^{-3} , batch size 64, with early stopping (patience 20 epochs) on the validation loss.

11.2. Training Diagnostics

Figure 18 shows the training and validation loss curves. Early stopping fires at epoch 85, with a train-validation gap of $\sim 10\%$, indicating moderate capacity limitation rather than severe overfitting.

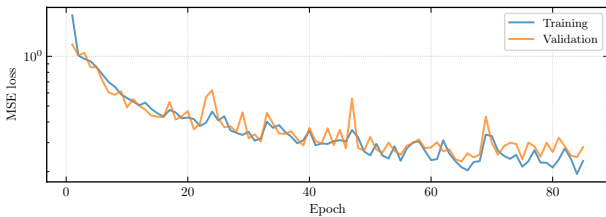


Fig. 18.— Neural network training and validation MSE loss vs. epoch (log scale). Early stopping at epoch 85 prevents overfitting while the train-validation gap of $\sim 10\%$ indicates moderate underfitting due to limited capacity.

11.3. Label Recovery

Figure 19 shows the NN label recovery on the CV set, and Table 6 compares the Cannon and NN performance.

The Cannon outperforms the neural network on every label, with the NN scatter being $1.3\text{--}2.1\times$ larger. Four factors explain this:

1. **Bad-pixel handling:** The Cannon naturally downweights masked pixels through their inflated uncertainties in the WLS training and χ^2 fitting. The NN receives zero-filled flux at masked pixels, which may introduce systematic features that the network cannot distinguish from physical absorption.
2. **No per-pixel noise model:** The Cannon’s χ^2 fit accounts for heteroscedastic per-pixel uncertainties. The NN treats all pixels equally in the MSE loss, giving undue weight to noisy pixels.
3. **Data efficiency:** The Cannon has ~ 22 parameters per pixel (21 coefficients + 1 scatter), and each parameter is constrained by 942 training stars. The NN has 4.5M parameters trained on 942 spectra—a $\sim 5000:1$ parameter-to-datum ratio that makes overfitting a persistent risk even with early stopping.
4. **Physical inductive bias:** The second-order polynomial encodes the physically motivated expectation that spectral features are smooth, low-order functions of stellar labels. The NN has no such prior, and must learn this structure from data alone.

11.4. Architecture Exploration

We additionally explored three alternative architectures:

Table 7: CV scatter for alternative neural network architectures.

Label	Baseline	Narrow	Deep	Regularized
T_{eff} (K)	94.3	84.1	85.4	175.3
$\log g$ (dex)	0.265	0.244	0.239	0.387
[Fe/H] (dex)	0.087	0.082	0.083	0.144
[Mg/Fe] (dex)	0.072	0.069	0.068	0.080
[Si/Fe] (dex)	0.058	0.056	0.054	0.062

The narrow MLP ($8575 \rightarrow 256 \rightarrow 128 \rightarrow 5$) performs best among NN variants, achieving $\sim 10\text{--}20\%$ improvement over the baseline. This is consistent with the hypothesis that the 4.5M-parameter baseline is over-parameterized for 942 training spectra. Adding explicit regularization (dropout + weight decay) performs worst, likely because the already-capacity-limited network cannot afford the additional reduction in effective capacity. Architecture tuning yields modest improvements, but The Cannon’s structural advantage of $\sim 2\times$ is not closed by any NN variant tested.

11.5. Advantages and Disadvantages

The Cannon excels in the low-data regime of this lab ($N \sim 1000$ training stars): its physically mo-

Table 6: Performance comparison between The Cannon and the neural network on the cross-validation set.

Label	Cannon bias	Cannon scatter	Cannon MAE	NN bias	NN scatter	NN MAE
T_{eff} (K)	+0.8	62.4	24.3	+13.5	94.3	48.3
$\log g$ (dex)	−0.004	0.128	0.062	−0.002	0.265	0.144
[Fe/H] (dex)	−0.004	0.045	0.018	+0.000	0.087	0.049
[Mg/Fe] (dex)	+0.008	0.045	0.023	+0.000	0.072	0.039
[Si/Fe] (dex)	+0.001	0.044	0.020	−0.002	0.058	0.025

Table 8: Qualitative comparison of The Cannon and neural network approaches.

Criterion	The Cannon	Neural Network
Interpretability	High: gradient spectra, per-pixel coefficients	Low: 4.5M opaque weights
Uncertainty model	Explicit: per-pixel scatter + measurement error	None: MSE loss only
Bad-pixel handling	Natural: inverse-variance weighting	Ad hoc: zero-filling
Flexibility	Limited to 2nd-order polynomial	Arbitrary nonlinear mapping
Training speed	~1 min (WLS per pixel)	~1 hr (GPU-accelerated)
Inference speed	~1 s/star (optimization)	~1 ms/star (forward pass)
Data efficiency	22 params/pixel, well-constrained	4.5M params, data-starved
Scalability	Fixed polynomial order	Can absorb more data, labels

tivated polynomial basis provides a strong inductive prior that compensates for limited training data. The neural network would likely improve substantially with larger training sets ($N \gtrsim 10,000$), where its capacity to learn nonlinear features would become an advantage. For production surveys, hybrid approaches such as The Payne (Y.-S. Ting et al. 2019) and astroNN (H. W. Leung & J. Bovy 2019) combine the interpretability of physics-based models with the flexibility of neural networks.

12. Areas for Improvement

Several improvements could enhance the model’s performance and reliability:

Higher-order polynomial models. The current second-order polynomial becomes inadequate at strong, saturated absorption lines. A third-order model would add $\binom{5+3-1}{3} = 35$ cubic terms per pixel (56 total), potentially improving the fit at the cost of increased risk of overfitting with only 942 training stars. Regularization (L2 or censoring as in The Cannon 2; A. R. Casey et al. 2016) would be essential.

Simultaneous continuum normalization. Training the continuum as part of the model (rather than as a preprocessing step) would eliminate the systematic errors introduced by imperfect normalization, which dominate the pixel-to-pixel correlations that inflate the MCMC uncertainty underestimate.

Correlated noise model. Modeling the pixel-to-pixel covariance (from the line spread function, continuum errors, and model inadequacy) would produce more realistic uncertainty estimates from the MCMC fit, closing the $\sim 10\text{--}15\times$ gap between formal and empirical uncertainties.

Larger and more diverse training sets. The current training set spans only four APOGEE fields. Expanding to a larger, more diverse sample would improve coverage of the label space edges (particularly

the warm T_{eff} boundary) and reduce the CV scatter.

Additional labels. Extending the model to include [C/Fe], [N/Fe], [Al/Fe], and other elements measured by ASPCAP would increase the astrophysical utility of the model. Each additional label increases the number of polynomial terms, so careful regularization would be required.

Neural network improvements. Incorporating per-pixel uncertainties into the loss function (e.g., heteroscedastic Gaussian likelihood loss), using learned embeddings rather than zero-filling for masked pixels, and employing convolutional architectures that respect the spatial structure of spectra would likely improve NN performance substantially.

13. Conclusions

We have constructed a data-driven spectral model for APOGEE DR17 red giant stars following The Cannon methodology (M. Ness et al. 2015). Starting from 1886 stars across four APOGEE fields, we trained a second-order polynomial model with 188,650 free parameters that achieves a training $\chi_r^2 = 0.998$.

Cross-validation on 944 held-out stars demonstrates precisions of 62 K in T_{eff} , 0.13 dex in $\log g$, 0.045 dex in [Fe/H], 0.045 dex in [Mg/Fe], and 0.044 dex in [Si/Fe], with negligible biases. Gradient spectra confirm that the model’s learned features align with known atomic and molecular absorption lines, providing physical validation of the data-driven approach.

A Kiel diagram constructed from the Cannon-fitted labels recovers the expected stellar-evolutionary morphology—the red giant branch, the red clump, and the metallicity gradient—in good agreement with MIST isochrone predictions. MCMC fitting of a mystery spectrum identifies it as a core-helium-burning, alpha-enhanced ([Mg/Fe] = +0.31, [Si/Fe] = +0.21) red-clump giant at sub-solar metallicity ([Fe/H] =

−0.61), consistent with a thick-disk or inner-halo origin. The formal MCMC uncertainties are $\sim 10\text{--}15\times$ smaller than the empirical CV scatter, highlighting the importance of unmodelled pixel-to-pixel correlations in spectroscopic uncertainty budgets.

Synthetic spectra illustrate that metallicity changes produce localized line-strength variations, while RGB evolutionary changes produce coupled temperature, pressure, and molecular effects across the full spectral range. These distinct spectral “fingerprints” allow the model to break the temperature-metallicity degeneracy. The effects of unresolved binary contamination on spectral label inference are discussed, with biases toward intermediate temperatures and diluted abundances.

A neural network comparison demonstrates that The Cannon’s physically motivated polynomial basis provides a strong inductive prior in the data-limited regime ($N \sim 1000$), outperforming a 4.5M-parameter MLP by $\sim 1.5\text{--}2\times$ on all labels. Future work could combine the strengths of both approaches through hybrid physics-informed neural networks.

Acknowledgements

Special thanks to Prof. Joshua Bloom and GSI Saahit Mogan for instructional support. I also acknowledge the use of ChatGPT Codex and Claude Code for assistance with vetting phrasing and checking the report writeup.

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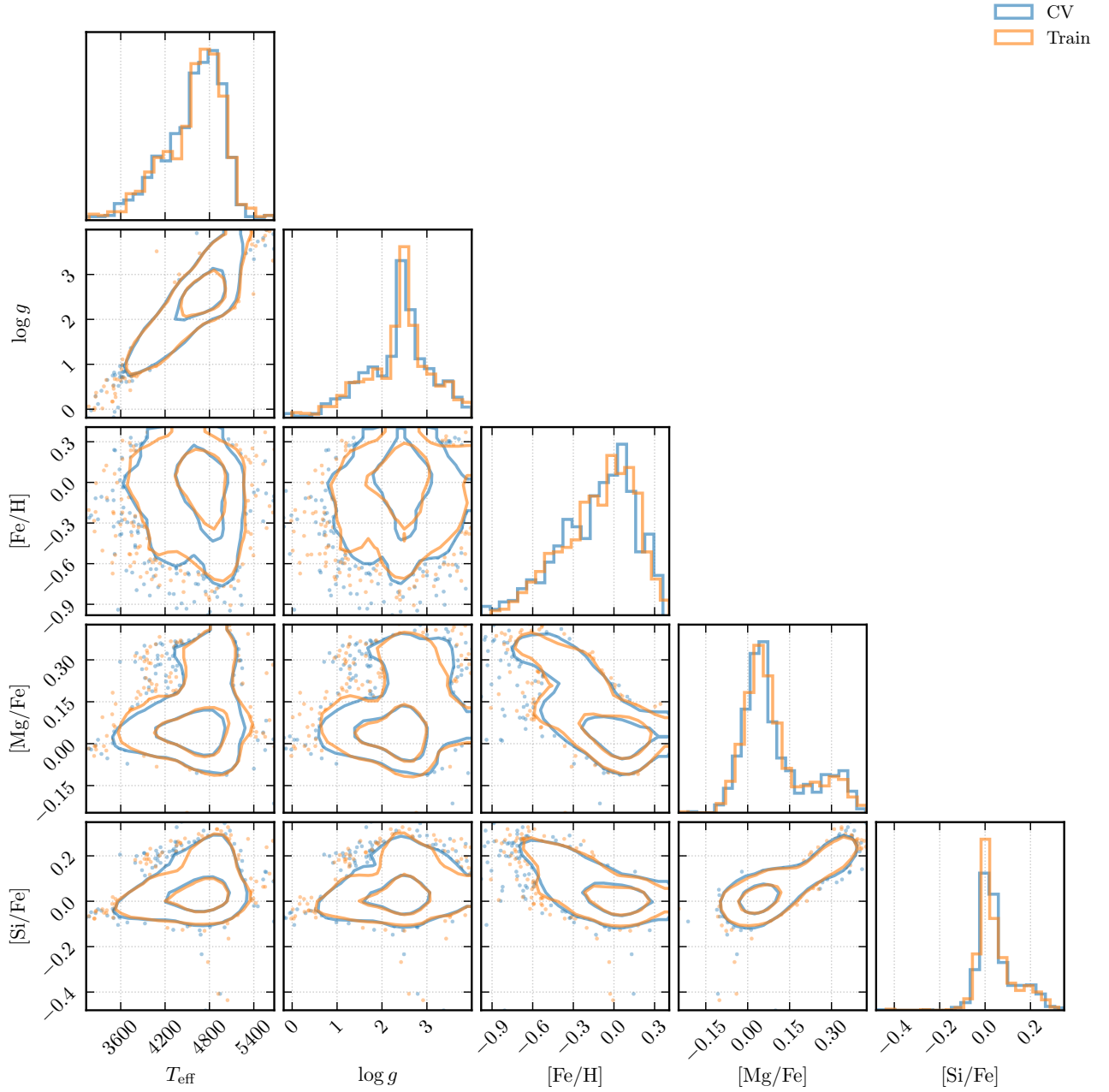


Fig. 7.— Corner plot of stellar labels colored by subset assignment (training vs. cross-validation). The stratified split preserves the label distributions of all four APOGEE fields in both subsets.

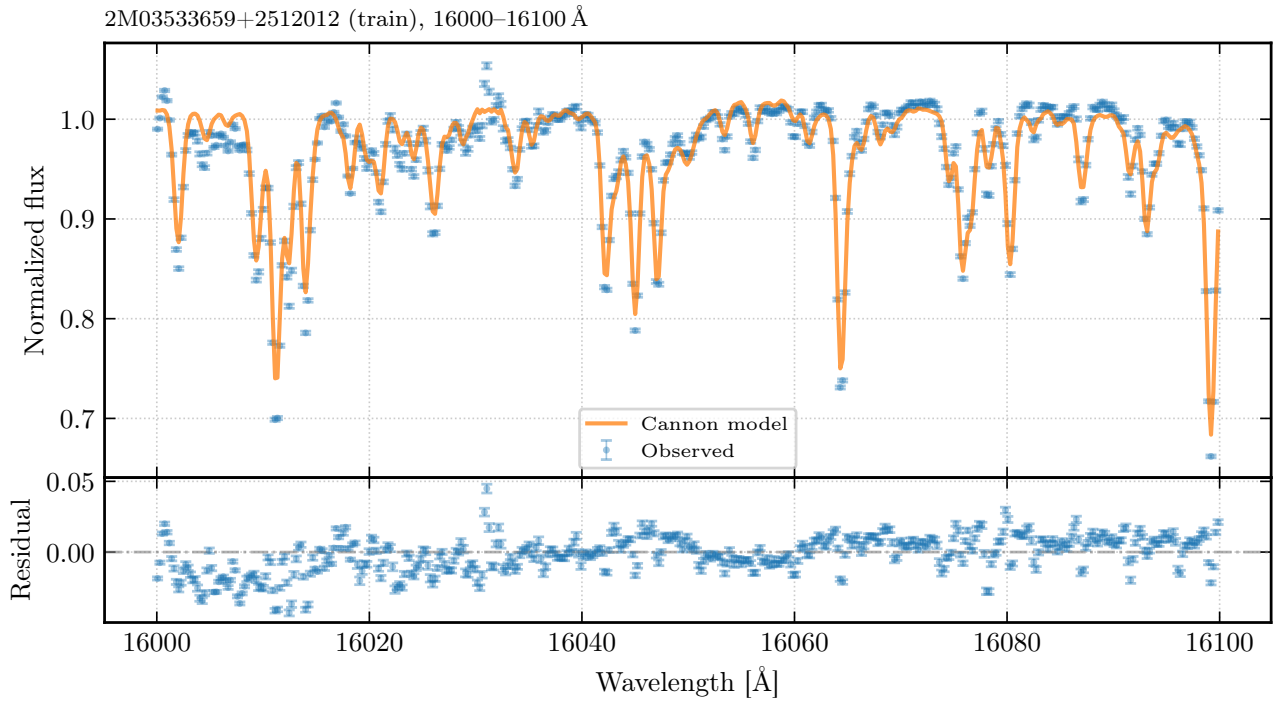


Fig. 8.— Training set prediction verification for star 2M03533659+2512012 in the wavelength range 16000–16100 Å. *Top*: Observed spectrum (black) and Cannon-predicted spectrum (orange) are nearly indistinguishable. *Bottom*: Residuals ($f_{\text{obs}} - f_{\text{model}}$) scatter around zero with standard deviation 0.013, comparable to the measurement uncertainty.

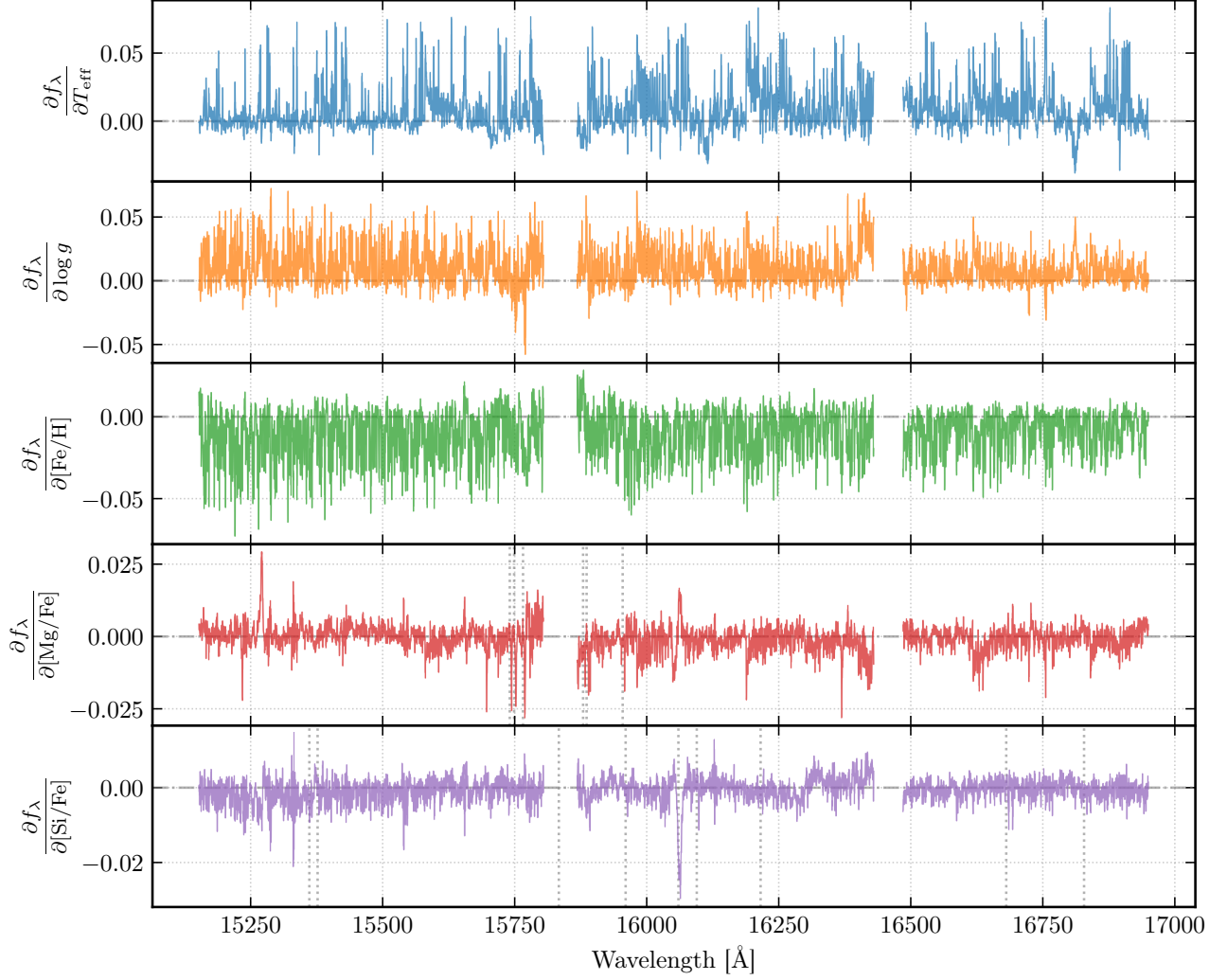


Fig. 9.— Gradient spectra $\partial f_\lambda / \partial \ell_i$ for each of the five labels. T_{eff} (top) shows a broadband response across the entire wavelength range, with the strongest features at CO molecular bandheads (>15580 Å). $\log g$ shows pressure-broadened features. $[\text{Fe}/\text{H}]$, $[\text{Mg}/\text{Fe}]$, and $[\text{Si}/\text{Fe}]$ show line-localized responses at known absorption features. Vertical markers on the $[\text{Mg}/\text{Fe}]$ panel indicate known Mg I lines (15741, 15749, 15765, 15879, 15886, 15954 Å; V. V. Smith et al. 2013); on the $[\text{Si}/\text{Fe}]$ panel, known Si I lines (15361, 15376, 15557, 15888, 15960, 16060, 16094, 16215, 16828 Å).

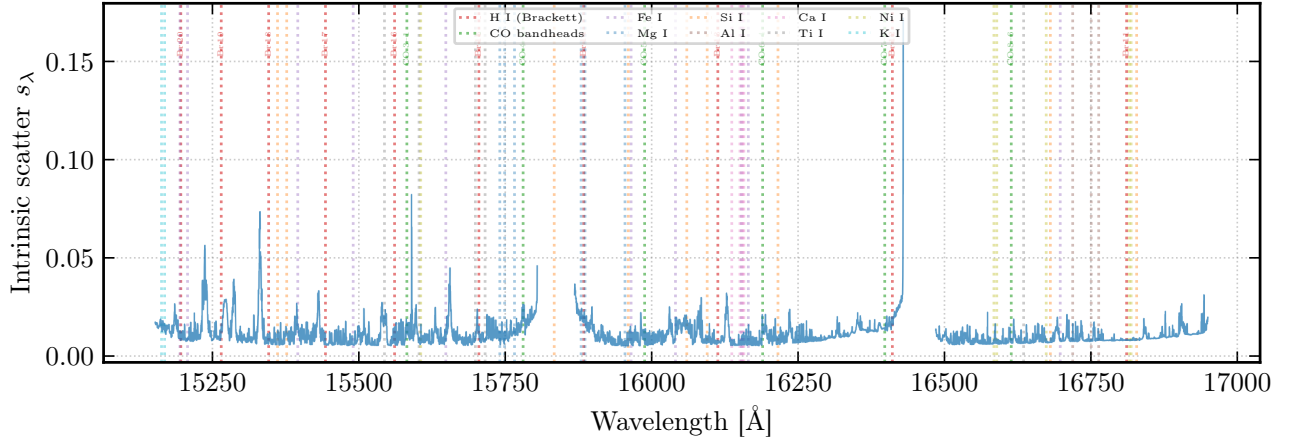


Fig. 10.— Per-pixel intrinsic scatter s_{λ}^2 across the APOGEE wavelength range, with known absorption line positions indicated (Brackett series, CO bandheads, and atomic lines from Fe I, Mg I, Si I, Al I, Ca I, Ti I, Ni I, K I). High-scatter regions correspond to strong absorption lines where the second-order polynomial is insufficient to capture the full label dependence. Chip-edge artifacts also contribute elevated scatter at chip boundaries.

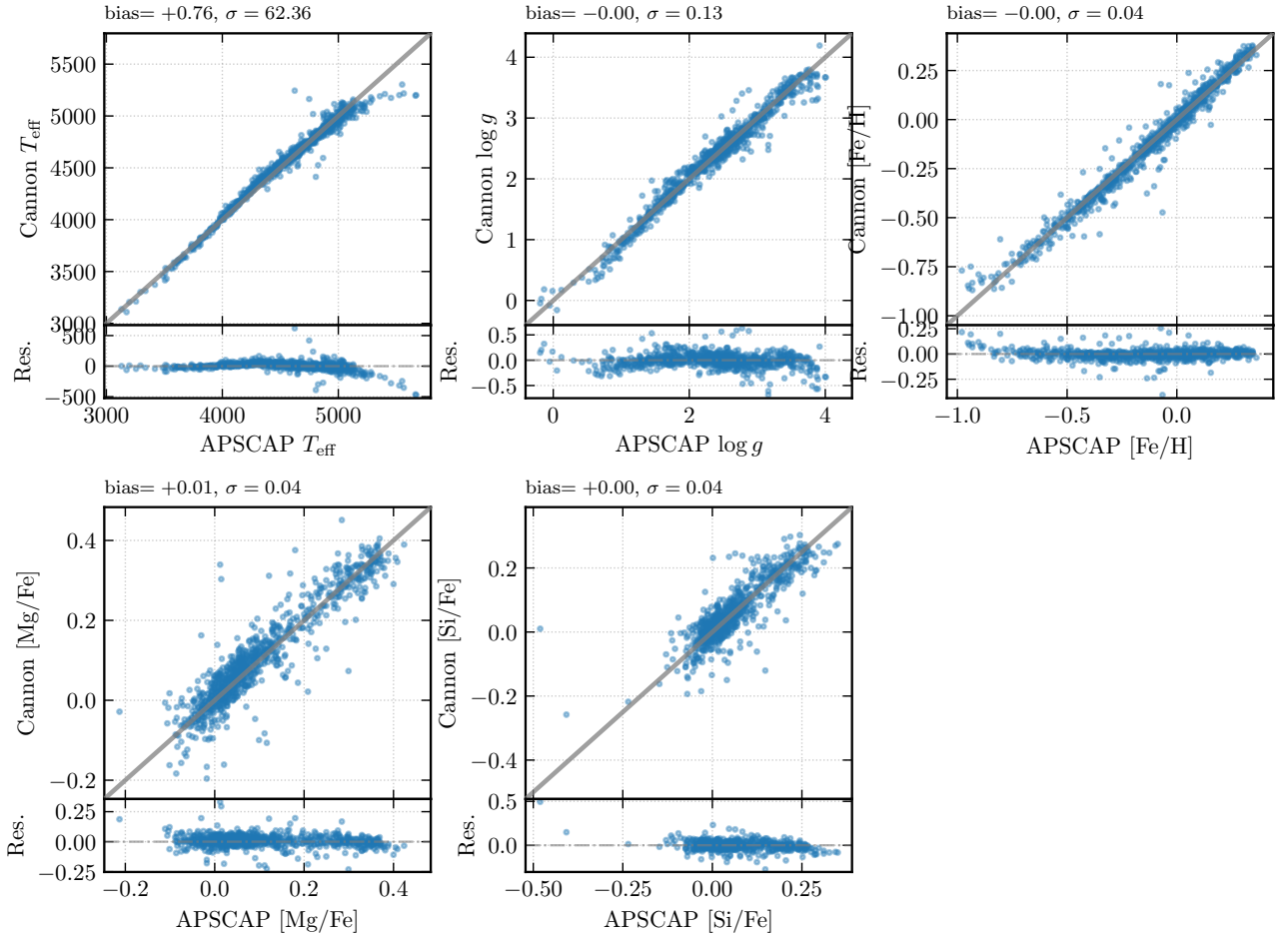


Fig. 11.— Cross-validation label recovery for 944 held-out stars. Each panel shows the Cannon-fitted label vs. the ASPCAP reference value, with a 1:1 line (dashed) and a residual sub-panel below. The three abundance labels ($[\text{Fe}/\text{H}]$, $[\text{Mg}/\text{Fe}]$, $[\text{Si}/\text{Fe}]$) are recovered with ~ 0.04 – 0.05 dex scatter; $\log g$ at 0.13 dex; T_{eff} at 62 K scatter (the largest in absolute units but smallest in fractional terms, $\sim 1.3\%$).

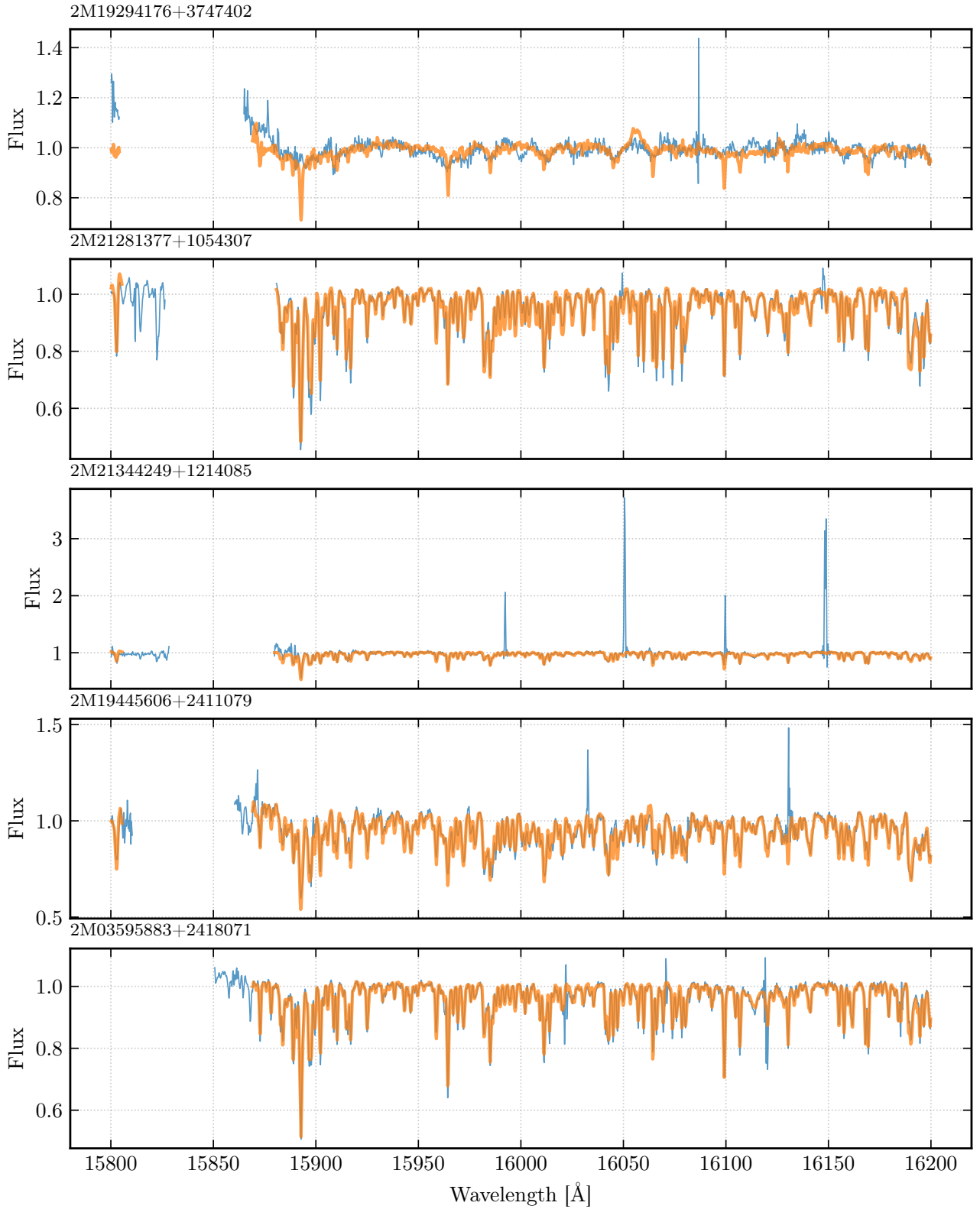


Fig. 12.— Observed (black) vs. Cannon-predicted (orange) spectra for the five worst-fit stars in the cross-validation set, shown in the 15800–16200 Å window. Systematic residuals indicate regions where the second-order polynomial model fails to capture the true spectral morphology.

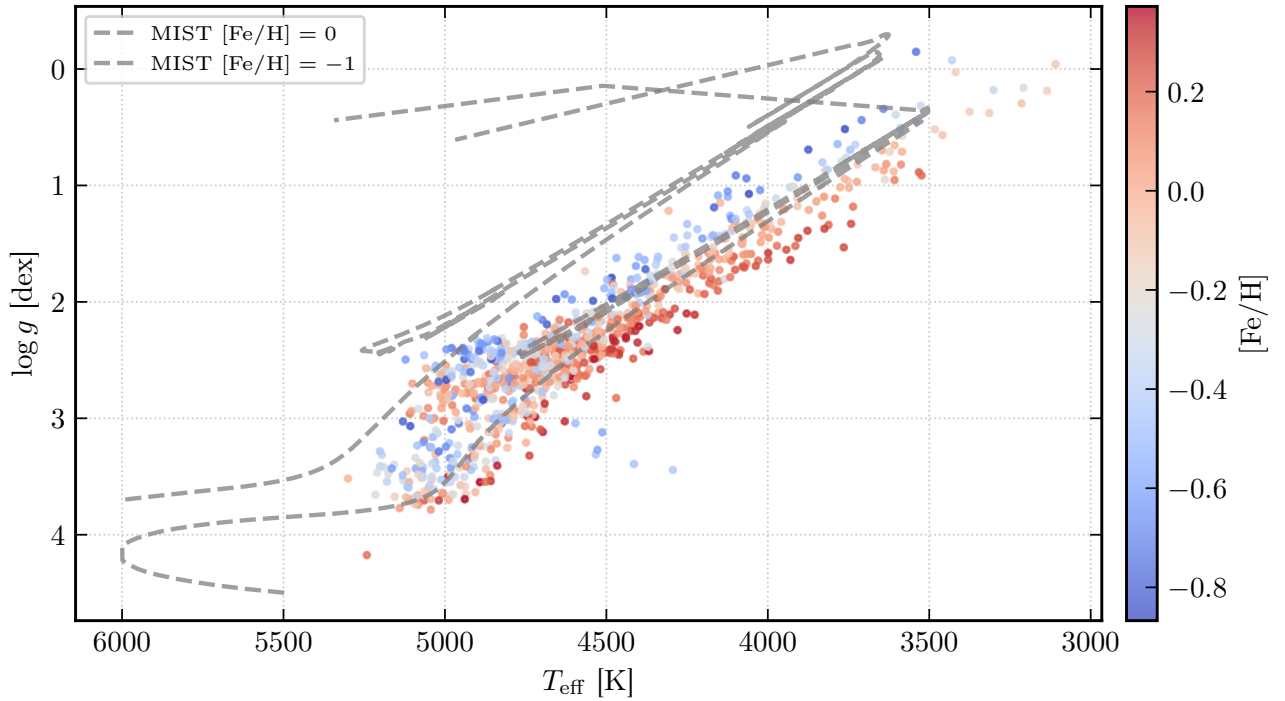


Fig. 13.— Kiel diagram ($\log g$ vs. T_{eff}) for 944 cross-validation stars, colored by Cannon-fitted $[\text{Fe}/\text{H}]$. Overplotted are 6 Gyr MIST isochrones at $[\text{Fe}/\text{H}] = 0$ (solid) and $[\text{Fe}/\text{H}] = -1$ (dashed). The red giant branch runs diagonally from lower-right (hot, high $\log g$) to upper-left (cool, low $\log g$). The red clump is visible as a dense concentration at $T_{\text{eff}} \sim 4800$ K, $\log g \sim 2.5$. Metal-poor stars (blue points) are systematically hotter at fixed $\log g$, in agreement with the isochrone shift.

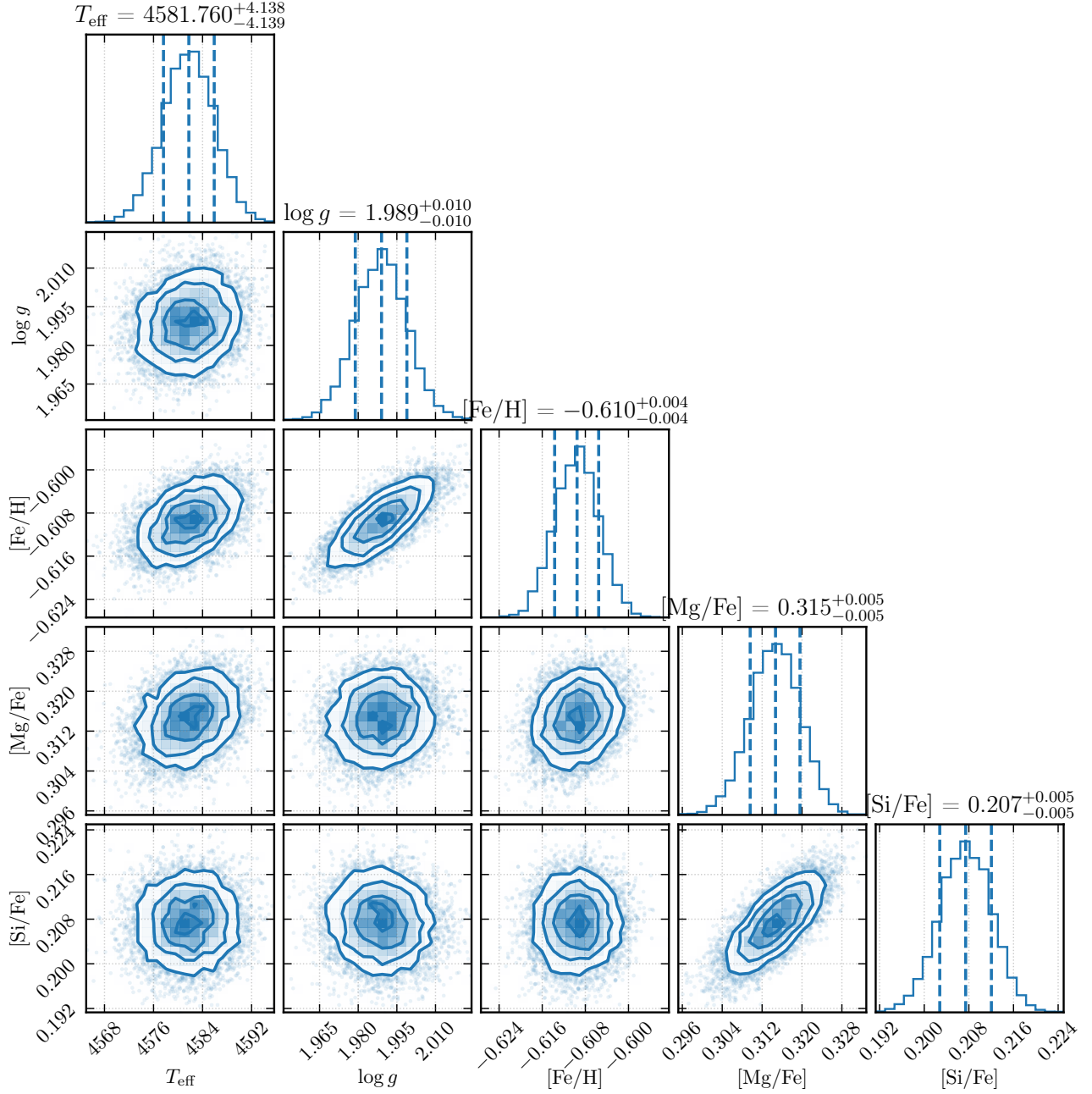


Fig. 14.— Corner plot of the MCMC posterior distribution for the mystery spectrum, showing the marginalized 1D and 2D projections of the five stellar labels. All parameters are tightly constrained, with the strongest correlation between T_{eff} and $\log g$.

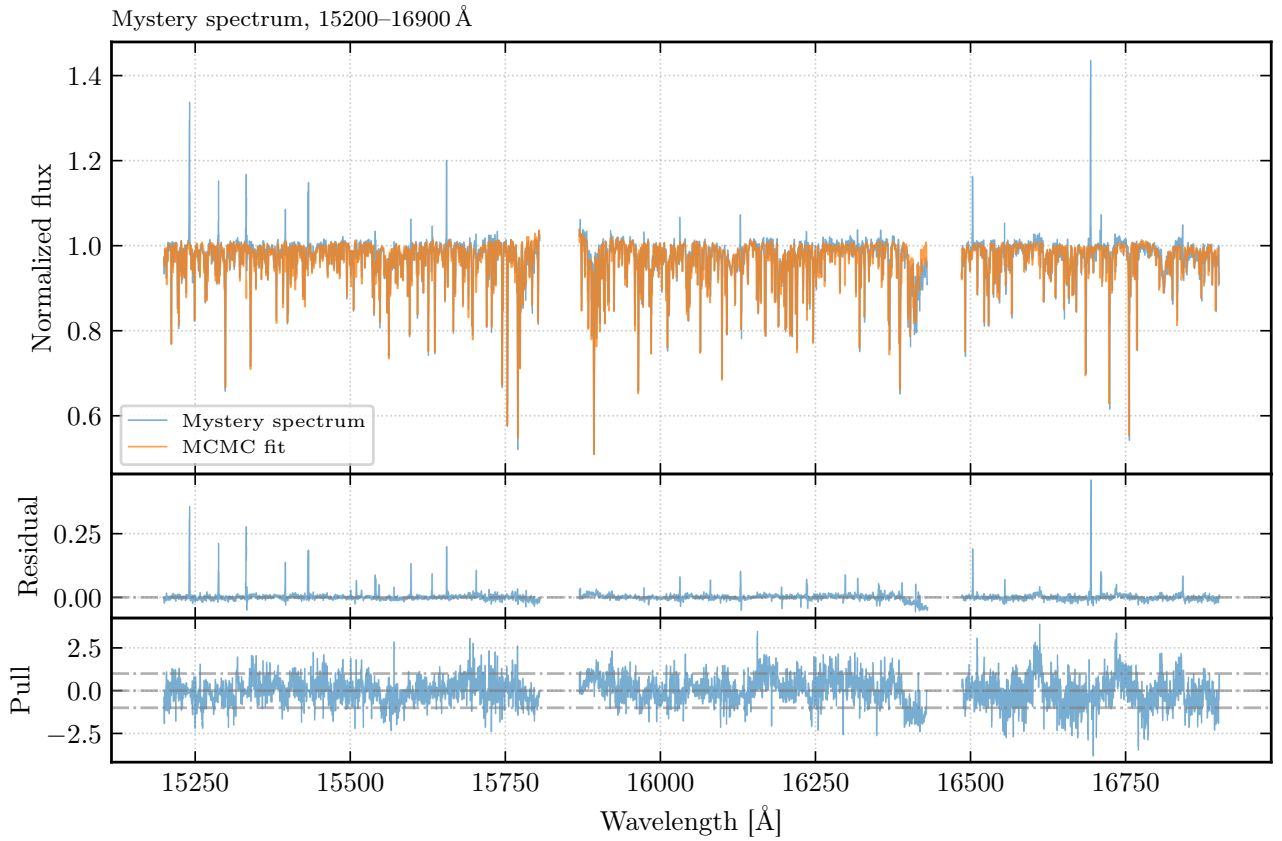


Fig. 15.— Mystery spectrum fit quality. *Top*: Observed normalized spectrum (grey) and MCMC-predicted spectrum at posterior median labels (orange). *Middle*: Residuals ($f_{\text{obs}} - f_{\text{model}}$). *Bottom*: Pull (residual divided by total uncertainty $\sqrt{\sigma^2 + s^2}$); the majority of pulls lie within ± 2 , confirming an adequate fit.

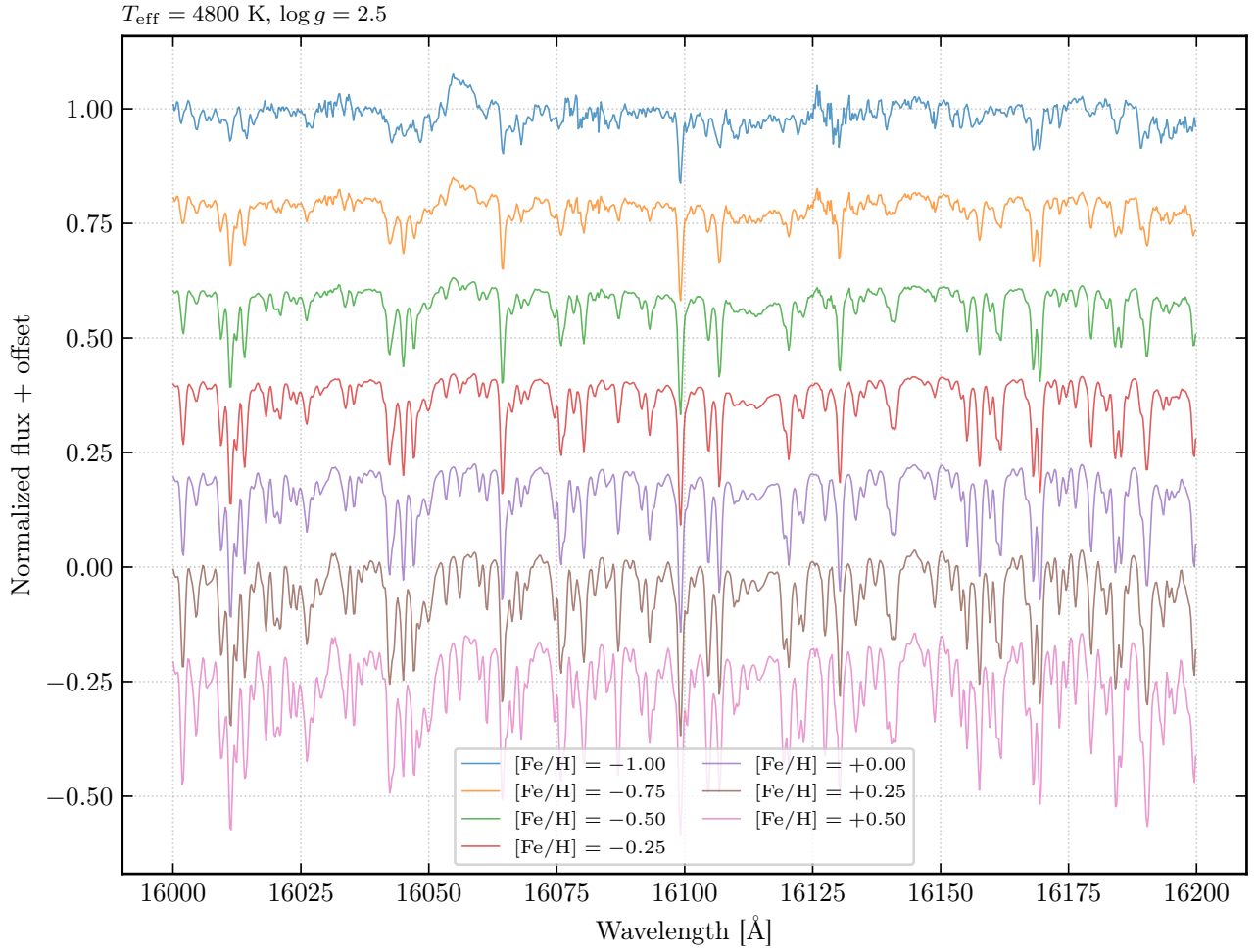


Fig. 16.— Synthetic spectra at varying metallicity ($[\text{Fe}/\text{H}] = -1.0$ to $+0.5$ in 0.25 dex steps), with fixed $T_{\text{eff}} = 4800 \text{ K}$, $\log g = 2.5$, $[\text{Mg}/\text{Fe}] = 0$, $[\text{Si}/\text{Fe}] = 0$. Spectra are vertically offset for clarity in the 16000–16200 Å range. Fe I absorption lines deepen monotonically with increasing $[\text{Fe}/\text{H}]$, consistent with the curve of growth. The continuum level also decreases at high metallicity due to line blanketing (the cumulative opacity of millions of weak lines).

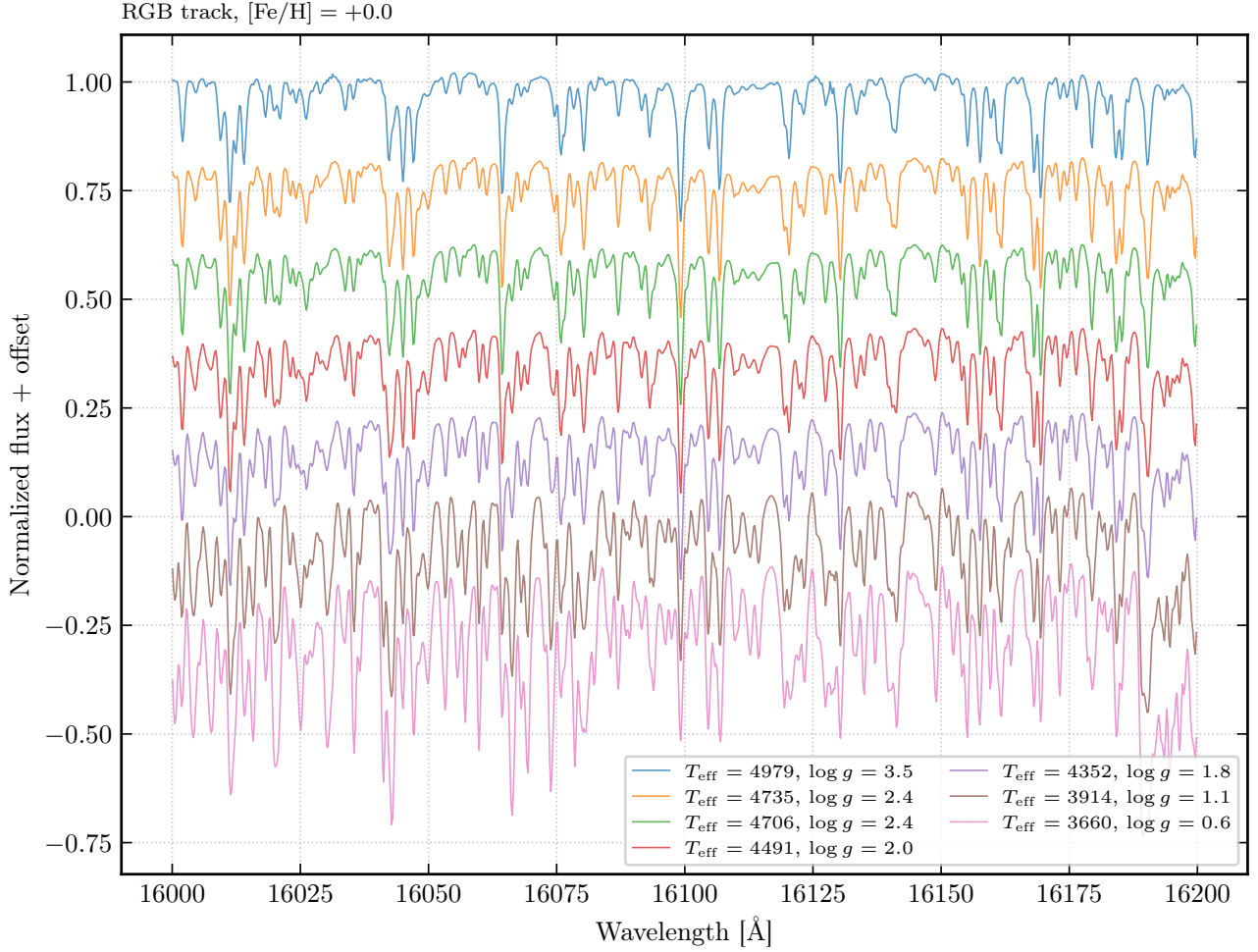


Fig. 17.— Synthetic spectra along the red giant branch at solar metallicity, sampled from a 6 Gyr MIST isochrone ($\log g$ from 3.5 to 0.5, with T_{eff} varying self-consistently). Spectra are vertically offset for clarity. As the star ascends the RGB (decreasing $\log g$, decreasing T_{eff}), absorption lines strengthen due to three coupled effects: Boltzmann/Saha ionization equilibrium shifts, pressure broadening weakens (lower $\log g$), and molecular bandheads (CO, OH) grow dramatically at low T_{eff} .

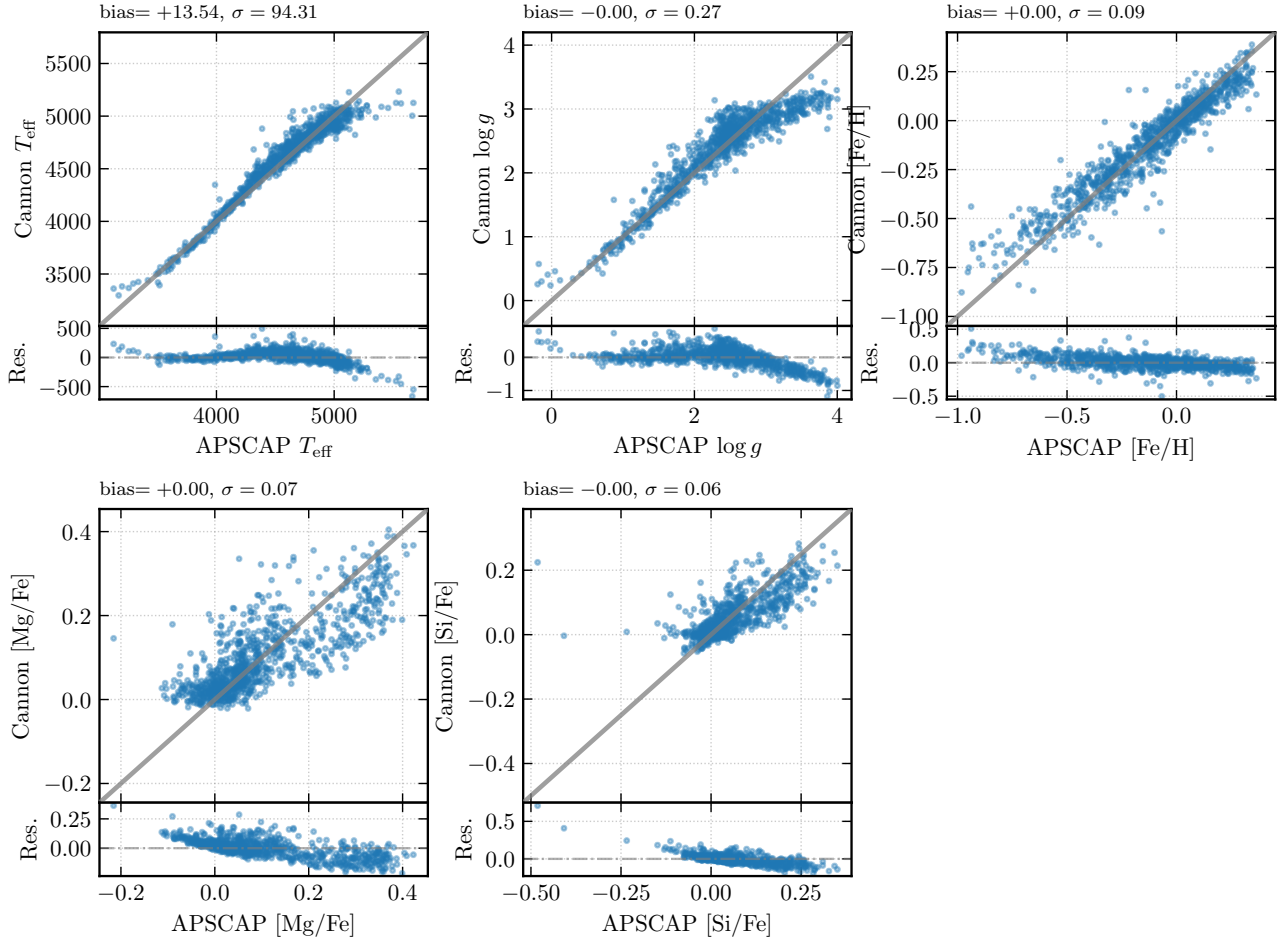


Fig. 19.— Neural network label recovery for 944 cross-validation stars, in the same format as Figure 11. Compared to The Cannon, the scatter is $\sim 1.5\text{--}2\times$ larger across all labels.

A. Reproducibility Notes

All code, notebooks, cached analysis products, and report sources used for this lab are version-controlled at <https://github.com/raw-asparagus/ay-128>.